Generation, Verification and Abstention in Memory-Constrained Financial Question Answering

Technical draft: 12 November 2026

Abstract

We investigate the interaction of evidence availability, executable program generation and selective verification in local financial question answering. A paired 160-question FinQA comparison finds 78 versus five reported-correct deliveries for a quantized 14B pipeline and a resident smaller F16 team. Later reference-informed constraints recover ten of 15 diagnostic cases, but an evidence-only planner on 96 TAT-QA contexts increases wrong deliveries and reduces correct candidates, failing its prespecified practical requirement. We analyze paired transitions, resource measurements, scoring conventions and unsuccessful replications. The findings concern specific recipes and support no universal causal ranking of architectures.

1 Introduction

Reliable financial question answering requires more than a correct arithmetic interpreter. A system must choose the relevant source values, compose a calculation and decide whether to deliver it. This article examines these decisions jointly through two complementary empirical studies. We organize the evidence around construct validity, paired comparisons and failure mechanisms, while retaining the broader development sequence as a technical appendix. The claim is an empirical diagnosis of evaluated recipes rather than a new inference algorithm.

2 Related Work

Financial and program-based reasoning. FinMAN already decomposes financial QA into formula selection, extraction, calculation and lightweight verification [Liu et al., 2025]; DCRC learns evidence auditing and executable program synthesis [Chen et al., 2026]. Bench360 measures deployment quality/resource trade-offs across inference configurations [Stuhlmann et al., 2026]. We do not claim these components or the general need for resource-aware evaluation as novel, nor reproduce those systems under our ceilings. FinQA provides expert questions and annotated programs over financial reports [Chen et al., 2021]. TAT-QA combines tables and text with arithmetic and other answer types [Zhu et al., 2021]; our experiments cover numerical FinQA and arithmetic TAT-QA subsets, not full benchmark leaderboards. PAL separates program synthesis from execution [Gao et al., 2023]. Program of Thoughts evaluates executable reasoning on financial datasets, including FinQA and TAT-QA, and discusses semantic binding [Chen et al., 2023]. Our source-reference grammar restricts operand provenance, but does not establish that the selected operands or formula answer the question. We test this limitation rather than claim a new program-reasoning method.

Planning, collaboration and verification. Multiagent debate reports gains from repeated exchange [Du et al., 2024]; self-consistency samples multiple reasoning paths [Wang et al., 2023]. We do not replicate either protocol. Tran and Kiela [Tran and Kiela, 2026] compare single- and multiagent reasoning under matched thinking-token budgets. Our ceilings include generated tokens across stages but do not equalize actual tokens, parameter-weighted compute or FLOPs. MAST analyzes system-design, inter-agent and verification failures across broader frameworks [Cemri et al., 2025]. Our evidence is narrower: saved arithmetic programs permit direct inspection of candidate correctness, source binding and final delivery.

Self-Refine demonstrates iterative self-feedback [Madaan et al., 2023]; Huang et al. [Huang et al., 2024] find intrinsic reasoning correction unreliable without external feedback. CRITIC incorporates tool feedback [Gou et al.,

2024]. These results motivate controls for both positive and negative effects of review. In recent financial QA work, VeriFin independently authorizes facts and formulas using XBRL and symbolic verification [Hall et al., 2026]; our source-membership checks are substantially weaker. BudgetVerify learns which verification tier to invoke from offline correctness and cost outcomes [Jenq and Shen, 2026]. We neither reproduce those systems nor claim priority for verification, repair or budget allocation. Our diagnostic reference structures explicitly supply information unavailable to an ordinary learned planner.

Adaptation and selective delivery. LoRA and QLoRA provide established parameter-efficient adaptation mechanisms [Hu et al., 2021, Dettmers et al., 2023]; role adapters are experimental components, not our novelty. Selective prediction distinguishes coverage from error among accepted predictions [Geifman and El-Yaniv, 2019]. Accordingly, we report correct, wrong and withheld outcomes over all questions. A lower wrong-delivery rate alone is not an improvement, and zero observed errors on a small constructed challenge is not a safety guarantee.

3 Methods and Evaluation

3.1 Programs, Candidates and Delivery

All models belong to the Qwen2.5 instruction-tuned family [Qwen Team, 2024]. Public evidence packets are built without answer labels. FinQA packets use lexical top-eight source-unit retrieval. The source-quantity catalog maps reference IDs to numerical values and source spans. Program generation uses constrained JSON with arithmetic operations (addition, subtraction, multiplication and division), supported constants, and references to earlier steps. A deterministic executor checks source membership and evaluates the final step. This provides arithmetic and provenance checks, not financial semantic validation.

Let K_i indicate a correct executable candidate and D_i indicate delivery on question i. The main rates are

$$C = n^{-1} \sum_i D_i K_i, \tag{1}$$

$$W = n^{-1} \sum_i D_i (1 - K_i), \tag{2}$$

$$H = n^{-1} \sum_i (1 - D_i), \qquad C + W + H = 1. \tag{3}$$

For repair policies, K_i refers to the final candidate; initial candidates are reported separately. A gate with no repair cannot deliver more correct answers than exist in its candidate pool. Candidate and delivery accuracy therefore answer different questions. Invalid outputs, refusals and rejected programs remain in denominators as withheld. Conditional error $W/(C + W)$ is descriptive of each policy's selected answers, not a matched comparison at fixed coverage.

3.2 Study A: Nominal 16 GB Deployment Recipes

The frozen cohort contains 160 FinQA development-source numerical questions from 50 company-year reports and 36 companies. Reports were selected by fixed ID hash after excluding recorded earlier report exposure; every eligible question in a selected report was retained. Public packets and scoring annotations are separate. Public-data and checkpoint pretraining overlap are not ruled out.

The single-agent system (SAS) uses an unadapted Qwen2.5-14B-Instruct Q4_K_M model for the pipeline. The multiagent system (MAS) keeps distinct unadapted 1.5B F16 extractor, 3B F16 drafter and 0.5B F16 verifier instances resident. The extractor suggests sources; the program and its execution are checked before delivery. This comparison changes parameter allocation, precision and role specialization together. "SAS" describes one model instance, not one model call.

Both recipes use CPU-only inference, eight threads, context 4,096, greedy decoding, seed 42, no memory mapping, and a 2,048 generated-token ceiling per question across stages. The primary profile includes a 128-token analysis allowance, two fixed training demonstrations, binary review and no repair. The direct control removes both analysis and demonstrations; the repair ablation adds full regeneration and review. These are bundled profile contrasts. Equal ceilings do not imply equal expenditure.

The completed Ubuntu run uses the frozen original code and pinned `llama-cpp-python` 0.3.16 runtime. It retains 16 worker attempts across interruptions, with an 11.5 decimal GB sampled controller-plus-worker RSS ceiling and a scope configured with `MemorySwapMax=0`. The Linux host reports approximately 15.44 GB usable physical RAM and a Ryzen 5 8645HS CPU. The historical run had process-swap failures; its resource invalidity is retained. The Ubuntu run repeats the same questions and is not a new statistical sample.

3.3 Study B: Diagnostics and Evidence-Only Planning

Windows comparisons use native `llama.cpp` b12202 CPU inference on a Ryzen 5 8645HS, generally context 4,096 and eight threads. Later runs impose a 6.1 GB aggregate worker Job Object commit ceiling in addition to the sampled 6.4 GB working-set guard. Earlier sampled-only pilots are not retroactively hard-capped. Commit, resident memory and paging are distinct quantities.

Exploratory interventions examine retrieval, catalog coverage, Q4/Q8 precision from a common F16 source, endpoint selection, model capacity, output format and review. Cohorts and their reuse are enumerated in the supplement. A 15-report enriched FinQA diagnostic requires that the annotated program be executable from the original catalog. Its two-by-two intervention supplies either the reference operand-value shortlist, reference operation scaffold, both, or neither. The scaffold fixes operations, constants, operand types, dependencies and endpoint. It is a substantial answer-key intervention; the shortlist can exclude valid alternatives and does not resolve equal-valued cell identities.

The deployable follow-up predicts operations and dependency edges from the question and evidence only. Its grammar uses unbound `VALUE` slots; a binding pass chooses actual source references and constants. Empty, invalid or disconnected shapes fall back to ordinary drafting. A 12-report development pilot motivated this revised grammar, which was frozen before held-out inference. No reference program or answer is supplied to this planner.

The main held-out cohort comprises 96 arithmetic table contexts from official TAT-QA development data, held out within this project. The secondary cohort comprises all 28 remaining eligible contexts from the official test data. Selection uses fixed hash ordering, one question per table, a nonempty numerical catalog, exclusions of prior table IDs, and normalized exact-table deduplication. Development contexts also exclude exact tables in the full test split. The source does not provide company-year mappings; report overlap across contexts and across benchmarks remains unverified. FinQA had only one eligible report left after recorded evaluation and actual adapter-training exclusions, motivating this benchmark change.

Four arms share evidence and a 960-token generation ceiling. Direct 7B emits a number and scale; program 7B emits a cited program and scale. Both use the same unadapted 7B Q4 checkpoint. Current MAS uses two separately resident 3B Q4 instances with fixed common-actor and control-critic adapters, allocating 768 tokens to drafting and 192 to review. Planner MAS allocates 192 to structure, 576 to binding (also the fallback cap), and 192 to the unchanged critic. Only executable candidates receiving PASS are delivered. This is a planner-plus-budget-allocation intervention, not planning at unchanged draft allowance. Both resident arms disable the RAM prompt cache.

The prespecified primary endpoint is planner-minus-MAS correct delivery. Practical success requires a positive paired interval *and no observed increase in wrong delivery*. The secondary cohort checks direction. The 7B arms are competitive comparators, not matched controls for the planner. Full templates, adapters, hashes and decoding details are in the reproducibility supplement and preserved traces.

3.4 Scoring and Statistical Analysis

Study A retains two historical numerical scores: reported-answer rounding with declared units, and raw execution-reference matching within $\max(10^{-5}, 10^{-5}|y|)$ of reference y. Neither is official FinQA program equivalence. Windows FinQA program experiments additionally translate arithmetic programs for the pinned official execution evaluator. Direct answers are compared with programs using their shared reported-answer metric, not an inapplicable program-equivalence score. TAT-QA uses the authors' pinned exact-match and F1 evaluator with predicted scales. Numerical label agreement does not establish unique or semantically correct provenance.

Study A intervals resample the 50 whole reports, retaining paired questions and question weighting, using 10,000 draws; company clustering is a sensitivity analysis. The paper's recomputation uses NumPy seed 42 and percentile quantiles. Its small endpoint differences from the original Python bootstrap are implementation differences, not changes in outcomes. Planner intervals reproduce the archived 10,000 paired table-context bootstrap draws (Python seed 73), alongside two-sided exact McNemar tests. Context pairs cannot substitute for unavailable report clusters. Tests for other interventions are exploratory; no multiplicity-adjusted discovery or equivalence claim is made. Bootstrap intervals

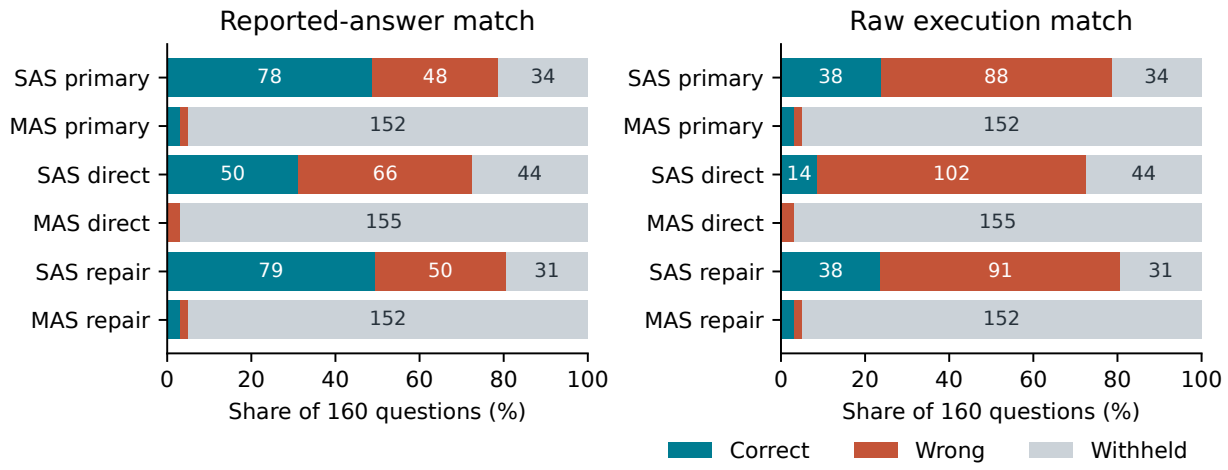


Figure 1: Study A: all six completed Ubuntu jobs, each on the same 160 questions. Segments show correct (teal), wrong (orange) and withheld (gray); labels are counts where space permits. Both scoring conventions are retained. Fewer MAS errors accompany substantially lower correct delivery.

reflect sampled cases, not seed or model-family variability. Sparse discordance can make percentile bootstrap intervals misleadingly reassuring.

4 Results

4.1 The Resident F16 Team Mostly Withholds

The primary SAS delivers 78 correct, 48 wrong and withholds 34 under reported-answer scoring; MAS delivers 5 correct, 3 wrong and withholds 152 (Figure 1). The MAS-minus-SAS correct-delivery difference is -45.6 percentage points, with a report-cluster 95% interval $[-53.9, -36.6]$. Raw execution matching changes SAS correct delivery to 38 and wrong delivery to 88, while MAS remains at 5 and 3. Thus the direction is robust to these scores, but the magnitude is not interchangeable across them.

Reported-score transitions are SAS-correct to MAS-correct/wrong/withheld: 5/1/72; SAS-wrong: 0/2/46; SAS-withheld: 0/0/34. Thus 46 avoided wrong deliveries accompany 72 lost correct deliveries through withholding, plus one correct-to-wrong transition. SAS has higher correct delivery on 33 reports, ties on 17, and loses on none. This is specific to reported scoring: raw scoring has one MAS-only correct answer. Conditional precision is 78/126 versus 5/8 (61.9% versus 62.5%), at 78.8% versus 5.0% coverage; similar precision does not establish equivalent usefulness. The other report-cluster effects are reported wrong -28.1 pp $[-37.1, -20.1]$, raw correct -20.6 $[-28.3, -13.4]$, and raw wrong -53.1 $[-61.9, -44.1]$. Company clustering and alternative weighting preserve these directions (supplement).

Candidate analysis locates the primary failure. SAS produces 150 executable candidates, 84 reported-correct; review withholds six correct and 18 executable wrong candidates. MAS produces only ten executable candidates, five reported-correct; it delivers all five correct and three wrong, withholding two executable wrong candidates. Its other 150 program-stage outputs are INSUFFICIENT, although all 160 responses parse as JSON. The team is therefore mainly generation-limited, not a strong solver hidden behind an excessively conservative final gate.

All annotated evidence spans occur in 138/160 packets. Within that reference-dependent stratum, SAS delivers 74 correct answers and MAS five; MAS still has 128 insufficient/invalid outputs. This descriptive audit does not prove unique catalog expressibility or causal sufficiency of retrieval. It does show that missing annotated spans alone cannot explain the observed coverage collapse.

For example, AAPL/2003/page_24.pdf-1 supplies 290 and 92 in the selected sentence. SAS subtracts them to obtain the reference 198; MAS selects the sentence but emits INSUFFICIENT. Conversely, on PNC/2018/page_171.pdf-1 the team passes a source-cited but numerically wrong calculation. These traces illustrate separate generation and verification failures; they do not isolate model capacity as their cause. The grammar lacks direct aggregate operations,

Constraint ($n = 15$)	Execution	Reported
Neither	0	0
Operand shortlist only	0	0
Operation structure only	4	4
Structure and shortlist	10	7

Table 1: Enriched FinQA diagnostic, correct candidates. The last three rows use reference information unavailable at deployment. No deployment accuracy claim follows from these scores.

and MAS withholds all 56 cases whose reference programs have multiple steps. Full traces and reference-defined strata appear in the supplement.

Direct control yields SAS 50/66/44 and MAS 0/5/155 correct/wrong/withheld. Repair yields SAS 79/50/31 and MAS 5/3/152. Initial candidates match the no-repair primary across paired cases. SAS repair releases one additional reported-correct and two wrong answers; by raw execution it releases zero correct and three wrong. MAS attempts 152 repairs without improving final delivery. More stages do not resolve the observed generation shortfall.

The Ubuntu run repeats all 960 historical delivery flags and score objects. Its 668,817 memory samples across 16 attempts show zero process-swap bytes and a maximum RSS of 10.946 GB, below the 11.5 GB sampled ceiling. This is stronger resource evidence than the earlier swap-affected run, but not uninterrupted execution or a bound on unsampled peaks. Primary SAS/MAS sampled peaks are 10.075/10.946 GB. Summed primary inference time is 11.668 hours for SAS and 3.383 for MAS; the latter's speed accompanies pervasive insufficient answers. Restart overhead and aborted work are excluded.

4.2 Evidence and Precision Do Not Resolve Generation

On 96 additional FinQA reports, an annotated correct program can execute from the original evidence/catalog on 82. Each matched 3B Q4 and Q8 condition misses 80 of those 82 and achieves only 2/96 official execution matches overall. The Q4/Q8 comparison has two gains and two losses, with a company-cluster difference interval about $[-4.3, +4.0]$ points. Increasing weight precision does not establish a general gain here; it is not evidence that precision never matters.

Wider lexical retrieval improves complete annotated-span coverage from 750/883 to 827/883 in a development screen. A 16-report generation pilot improves from 1 to 4 official matches, but a separate 21-report replication changes 2 to 1 (one gain, two losses). All three initial gains use only original spans. Expanding the catalog improves an enriched five-report diagnostic from 0 to 2, but changes 2/96 to 1/96 on the broader cohort. Retrieval coverage, catalog expressibility and generated accuracy are distinct outcomes.

The 15-report answer-key diagnostic shows where conditional potential remains (Table 1). Operand restriction alone leaves accuracy at zero despite reducing the median selectable IDs from 23 to two. Structure alone recovers four answers; structure plus the shortlist recovers ten official matches but seven reported matches. The remaining five official failures assign at least one source slot a different numerical value from the reference; three additional official-correct outputs mislabel a fractional result as percentage points. Because the scaffold supplies most of the calculation, this is not a learned planning result or an upper bound over all valid reasoning strategies.

4.3 The Evidence-Only Planner Fails Its Criterion

All four arms complete the 96-context cohort (Figure 2). Direct 7B delivers 19 correct, 76 wrong and withholds one; program 7B delivers 15, 80 and one. Current MAS delivers 6, 23 and 67; planner MAS delivers 10, 25 and 61. Planning gains four correct deliveries and loses none: $+4.2$ percentage points, paired context-bootstrap 95% interval $[+1.0, +8.3]$. However, only four pairs are discordant, giving exact $p = 0.125$. We do not treat the positive percentile interval as reliable evidence of a population gain. Wrong deliveries increase by two, so the prespecified practical criterion fails regardless of that interval.

Correct candidates decline from 26 to 25, with four gains and five losses. All four candidate gains are delivered. Two gains acquire a matching operator tree; the other two already had matching tree structure and improve numerical binding. Among 87 parseable reference derivations, operator-tree matches fall from 46 to 40 (three gains, nine losses). This diagnostic replaces numerical leaves with placeholders and sorts commutative arguments; it does not recognize all algebraic equivalences or verify source identities. Figure 3 reports uncertainty for these paired changes.

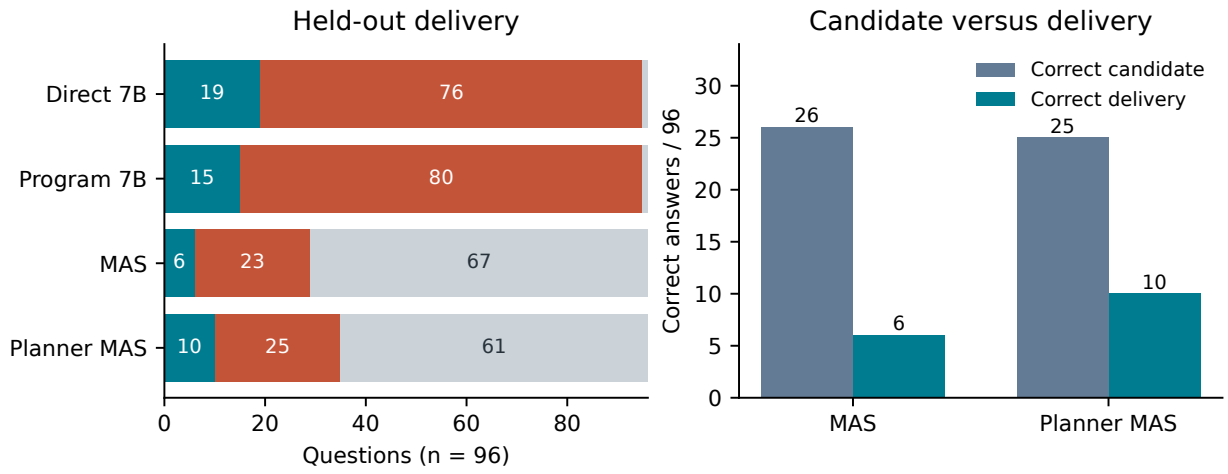


Figure 2: Primary held-out TAT-QA cohort. Left: all correct, wrong and withheld deliveries. Right: correct candidates before review versus correct final deliveries. Planning delivers more correct answers but generates fewer correct candidates and more wrong deliveries.

Connected shapes are usable on 57/96 contexts. The other 39 have operations outside the final dependency graph and fall back; all 39 saved drafts and verdicts reproduce the baseline. Passing structural validation does not guarantee a correct calculation. The fixed critic rejects 20 of 26 correct baseline candidates and 15 of 25 planner candidates. Its changed acceptance pattern improves correct delivery without a net generation gain and still accepts 25 wrong planned answers.

On the separate 28-context official-test cohort, both MAS arms deliver two correct, ten wrong and withhold 16; correct candidates decrease from five to three. This is a small negative directional check, not proof of equality. The 7B comparison stops after two contexts because of a 352,342,016-byte CPU compute-buffer allocation failure; a recovery attempt also fails. We retain the partial records but do not report a complete four-arm official-test comparison.

Planning also costs time: median resident request latency increases from 25.2 to 33.9 seconds; generated tokens increase from 6,672 to 9,863. Peak worker commit remains about 5.26 GB (Figure 4). Direct and program 7B medians are 34.3 and 46.8 seconds with approximately 5.44 GB peak commit, but include model loading and are not matched resident latency comparisons.

4.4 Stronger Baselines and Nontransferable Gains

The result is not that larger or direct models always fail to help. On 120 fresh TAT-QA contexts, direct 7B scores 29 versus 12 for direct 3B (22 gains, five losses; exact paired $p = 0.00151$). An earlier 40-context comparison is inconclusive at six versus five. On the shared 96-report FinQA cohort, fixed 7B direct output scores 35 versus 20 for cited programs under reported-answer matching (21 direct-only, six program-only; $p = 0.00592$). In contrast, a 72-context TAT-QA subset favors programs 19 versus 14 (14 program-only, nine direct-only; $p = 0.405$). Output-format effects depend on task, representation and scoring; no universal direct-answer advantage is supported.

Scoring sensitivity is substantial. FinQA's unmodified official execution score for those 7B programs is 1/96. A post-hoc declared-percent conversion recovers 18, yielding 19/96; the original score is retained. Three unsupported reported-reference labels remain in denominators. These observations do not license replacing the primary scorer with a favorable convention.

FinQA actor adaptation transfers to 72 other TAT-QA contexts with correct candidates increasing from three to eight (six gains, one loss; $p = 0.125$); the critic delivers six correct, ten wrong and withholds 56. On fixed FinQA candidates, an initially favorable trained-critic pilot does not reproduce zero wrong deliveries in a 21-report transfer: control delivers 1/5/15 and crossed critic 1/2/18. Earlier crossed-critic development also fails its no-correct-loss adoption rule against the stronger original critic. These observations remain part of the negative sequence.

The clearest engineering improvement is cache configuration. On a fixed resident 21-report workload, disabling RAM prompt caching preserves every draft, verdict and delivery while reducing peak worker commit from approx-

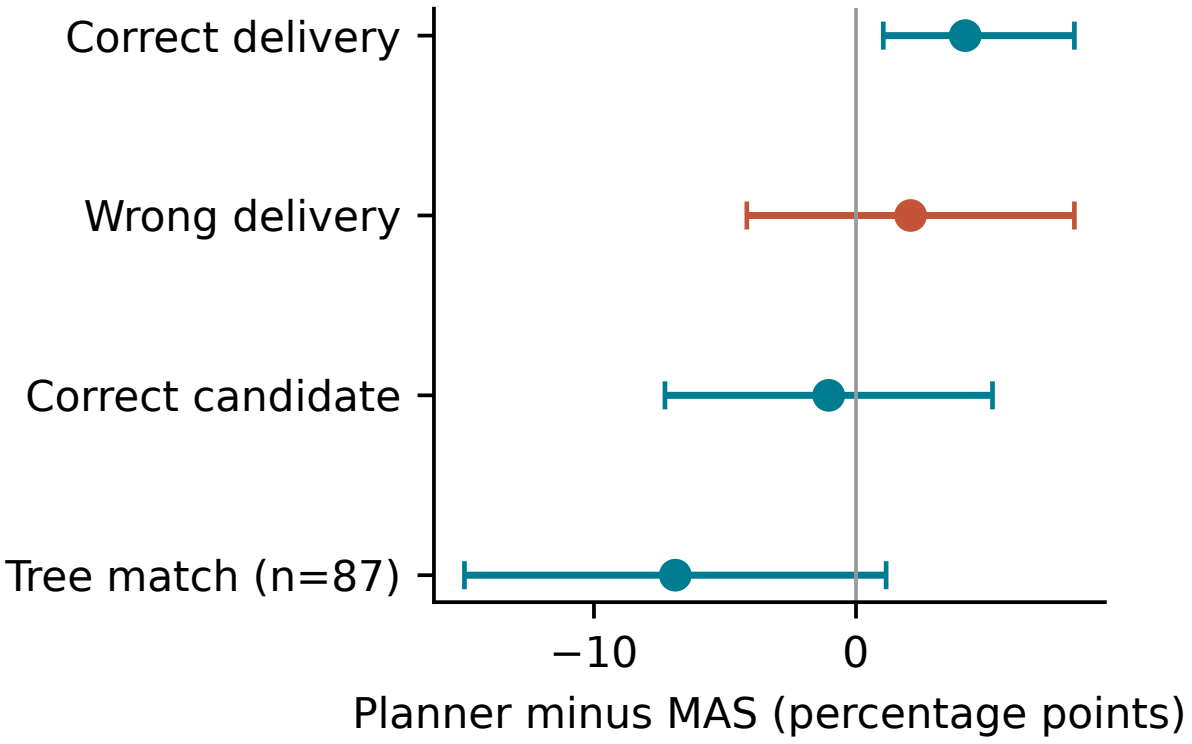


Figure 3: Planner-minus-MAS effects with archived 95% paired context-bootstrap intervals. Correct delivery has only four discordant pairs (exact $p = 0.125$). Wrong delivery is an adverse outcome; its observed increase fails the practical criterion. Tree agreement uses 87 parseable cases.

imately 6.101 to 5.264 GB and eliminating logged prompt-cache allocation failures. The default-cache completion involved allocation fallback near the cap. This change improves observed resource headroom, not reasoning accuracy, and does not prove absence of Windows paging.

5 Discussion and Limitations

The evidence supports a sequence of distinct constraints. In Study A, the selected F16 team generally does not attempt a useful calculation. In Study B, available reference-executable evidence can coexist with wrong candidate programs. Supplying reference structure exposes conditional potential, but predicting usable structure remains difficult, source assignment can still fail, and review can discard correct candidates or accept wrong ones. The planner's final delivery increase is therefore insufficient evidence of a generation improvement.

This is an empirical result about specific Qwen2.5 recipes, prompts, grammars and financial subsets. Parameter size and precision are confounded in Study A; the Windows planner changes draft allocation as well as adding a stage. Greedy single-seed experiments do not estimate training-seed or stochastic-decoding variance. Repeated development studies, enriched oracle diagnostics and reused cohorts cannot be pooled as independent confirmations. Project-held-out public data do not establish freedom from pretraining contamination; FinQA splits overlap at report level, and TAT-QA report clusters are unavailable. Sparse paired improvements and multiple exploratory comparisons limit inferential strength.

Resource claims are equally bounded. Windows job commit is not physical RAM feasibility; sampled RSS can miss peaks. Linux swap-disabled scope records and zero sampled process swap strengthen the Ubuntu evidence, but the run was interrupted and the SSD suffered filesystem errors. Seven unreadable analysis artifacts were reconstructed offline: all six replay hashes and all 40 recorded evidence hashes match, as does the regenerated summary; the regenerated text report has no original recorded hash. The original export tarball and one bootstrap log remain

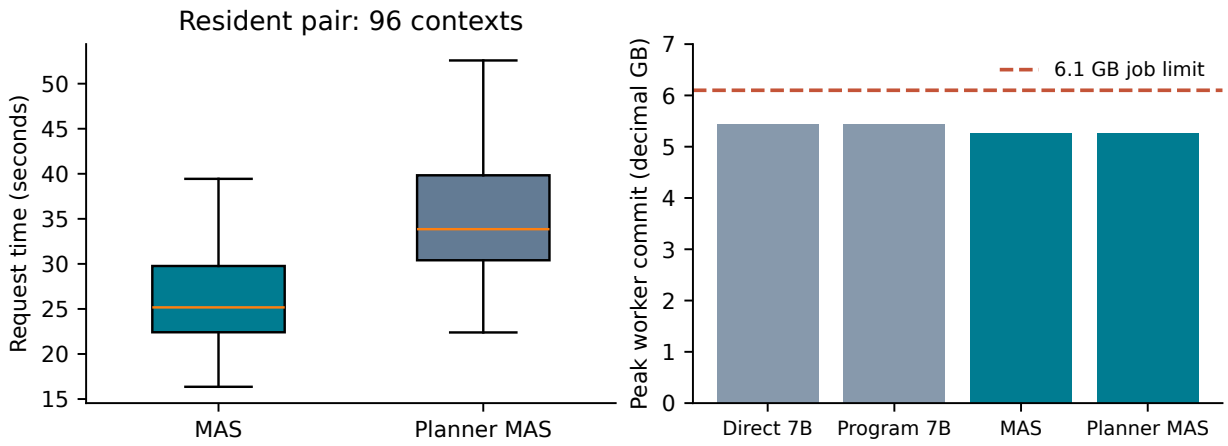


Figure 4: Primary held-out resources. Left: resident request-time distributions (median, quartiles and 1.5-IQR whiskers; outliers omitted from display, retained in data). Right: observed peak worker commit and configured limit. Peaks are descriptive measurements, not physical-RAM feasibility bounds; cross-arm loading policies differ.

unavailable. Separately verified repository archives preserve the usable evidence. These qualifications preclude claims of uninterrupted deployment reliability.

We have not evaluated frontier models, alternative families, a broad prompt search, trained planners, XBRL-authorized formulas, or all published verification baselines. We report numerical benchmark agreement rather than expert financial adjudication, calibration or operational safety. Full prompts may contain ambiguity, and correct numbers do not ensure correct provenance. The systems' error rates do not support autonomous financial decisions. The supplement retains historical positive, null and adverse findings, unsuccessful resource attempts and a machine-readable artifact census; no historical result is silently revised.

6 Conclusion

The observed errors cannot be summarized by one accuracy number. The F16 team loses useful coverage; reference-informed constraints reveal conditional recovery; evidence-only planning fails its practical requirement. Research on local financial reasoning should therefore measure candidate quality and delivery together, preserve scorer sensitivity, and treat resource feasibility as a separate empirical claim.

A Scope and Evidence Status

This supplement accompanies the main paper and retains the broader research sequence, including weak baselines, unsuccessful interventions, diagnostic successes, failures to transfer, and resource interruptions. It is organized by evidential role rather than treating every development experiment as a confirmatory trial. Repeated questions, repeated controls and post-hoc diagnostic outputs are not pooled into a sample-size claim. No neural inference was run to prepare either manuscript. Historical files remain unchanged.

The paper package includes an artifact inventory with hashes, an unfiltered collection of 222 result/score/analysis documents selected by filename from the research and preserved-evidence trees, independently regenerated headline metrics, and raw-score checks. The collection is a machine-readable census, not 222 independent experiments. The complete research notes and `DECISIONS.md` remain authoritative for chronology. Superseded experiments, failed training fits and incomplete jobs remain in the repository and archives. The historical manuscript is preserved separately and should not be used for the new Ubuntu resource measurements.

Verification levels. All 20 handoff archives and their 2,930 mapped source files pass recorded byte-length and SHA-256 checks. The broader inventory records 3,776 research and preserved files, including duplicated evidence

representations. Main Ubuntu results were recomputed from raw case ledgers: 1,920 initial/final programs were re-executed and rescored, all 960 records reconciled with the published export, and all stage-generated-token sums checked against the 2,048 ceiling. All 668,817 memory samples from 16 worker attempts pass the recorded RSS and process-swap checks. Fourteen Windows FinQA paired ledgers were re-executed and rescored using the pinned official evaluator and reported-answer scorer; seven TAT-QA prediction ledgers were rescored. The 96-context planner and 28-context secondary MAS outcomes were independently rescored using the official TAT-QA implementation. Older result documents are retained with their original audit provenance; the present checks do not claim independent expert adjudication of every financial label or a fresh replay of every historical training trajectory.

B Reproducibility Specification

B.1 Study A: Frozen 16 GB Recipes

The Ubuntu run identifier is `lab-16-ubuntu-noswap-v1`. The frozen protocol identifier is `finance-deployment-lab-v2-115gb`. The public corpus contains 160 questions from 50 company-year reports (36 companies), selected from FinQA development data by report hash after recorded prior-exposure exclusions. This is not official-test performance. Six jobs combine two arms with primary, direct-control and repair-ablation profiles. The paired unit is the original question, with report clustering; 960 outputs do not represent 960 independent questions.

Setting	Recorded configuration
Single model	Qwen2.5-14B-Instruct, GGUF Q4_K_M; no project adapter.
Resident team	Qwen2.5-1.5B-Instruct F16 source suggestion; 3B F16 program drafting; 0.5B F16 binary verification. Separate simultaneously loaded model instances, no shared backbone.
Inference	CPU only, 8 threads, context 4,096, greedy temperature zero, seed 42, GPU layers zero, `use_mmap=False`.
Output budget	2,048 generated tokens per case across all stages. Actual token use differs by arm/profile; prompt tokens sum every call, not one context window.
Primary	Analysis allowance 128; two fixed training demonstrations; binary review; no repair.
Direct control	Removes both analysis and demonstrations, retaining the structured program and review pipeline. It is not the direct-number Windows comparator.
Repair ablation	Same primary initial trajectory plus full regeneration/review after failure. Initial trajectories are matched; later candidates may change.
Evidence	Lexical IDF/length-normalized top-eight source units; compact quantity catalog; answer keys absent from generation packets. MAS receives source suggestions.
Runtime	`llama-cpp-python` 0.3.16, NumPy 2.5.4, psutil 7.2.2; Python 3.12.15; Linux 7.0.0-38-generic, glibc 2.43, x86-64.
Hardware	AMD Ryzen 5 8645HS; Linux physical RAM recorded as 15,438,376,960 bytes. GPU is not used for these inference runs.
Resource control	11,500,000,000-byte sampled controller-plus-worker RSS guard, process-swap monitor, and user scope reporting `MemorySwapMax=0`; not a hard RSS cgroup limit.
Resume	Original manifests and per-attempt traces retained across interruption/reboot. A one-page usable-RAM reporting drift was reconciled only with the remaining guarded identity fields matching.

Model-file and source SHA-256 values are preserved in `reproducibility/identities.json`; no large model tensors are bundled with the paper. The original source public-packet hash is `9611c2f650983b9cd8d04823824cbf1999d6bbb00fea9cea3c9894875cb9bf59`. Exact prompt text and responses for every original stage are in the restored case ledgers. Representative complete prompts, selected mechanically as the first recorded

case in each job, are included in `reproducibility/prompts/`. They are examples of recorded inputs, not replacements for the frozen templates or full case-specific evidence.

B.2 Windows 8 GB Application Budget

Windows 11 build 10.0.26200 runs on an AMD Ryzen 5 8645HS with 6 physical/12 logical CPUs and 16,390,729,728 physical bytes. The inference executable is `llama.cpp` b12202, revision `d6be06557`, x64 CPU. Its distribution archive hash is retained with the environment record. The reported environment includes Python 3.12.14 and psutil 7.2.2. Both resident workers use context 4,096 and eight CPU threads; execution of drafting/review stages is sequential while both weights remain loaded. The shared Job Object has a 6,100,000,000-byte commit limit, and the separate working-set guard is 6,400,000,000 bytes. The initial Windows pilots used sampled RSS only; tables and narratives do not upgrade their guard retrospectively.

The direct/program 7B comparator uses the two-shard Qwen2.5-7B-Instruct Q4_K_M checkpoint. Small-model conditions use the recorded Qwen2.5-3B-Instruct Q4 checkpoints; the matched Q4/Q8 study quantizes one shared F16 source and must not be conflated with comparisons using other pre-existing files. Role-adapted runs attach the common actor or control critic LoRA to the appropriate separate 3B instance. Artifact names, sizes and hashes distinguish these identities in the archived model and launch records. All paper generation is based on those identities; matching a parameter count alone is not sufficient to reproduce a condition.

For the held-out planner, temperature is zero and the common generation ceiling is 960. Current MAS reserves 768/192 for draft/review; planner MAS reserves 192/576/192 for shape/binding/review. A rejected shape still consumes its planning work and falls back within the 576-token drafting allowance. Direct/program 7B each reserve 960 without review. Their sequential CLI duration includes loading; resident pair request times exclude initial worker loading. The measured latency difference between the two resident arms is the cleanest planner contrast. Prompt RAM caching is disabled with `--cache-ram0`; context length is unchanged.

B.3 Adapter Provenance and Training

Adaptation is an established baseline component [Hu et al., 2021, Dettmers et al., 2023], not evidence of algorithmic novelty. Initial 3B adaptation uses an NF4 training base from `unsloth/Qwen2.5-3B-Instruct-bnb-4bit`, pinned revision `2672b588848527e456d320c11926c794539f47d5`. The deployment GGUF Q4_K_M base is a different quantization; exact full-precision weight equivalence is not asserted. Invalid source-binding corpus revision 01 was superseded and is not presented as usable training data.

The valid initial corpus has 761 examples fitting the full 1,024-token context constraint, split 661 train/100 validation by recorded report groups. The completed initial epoch uses 83 optimizer updates and 14,966,784 trainable parameters across 504 changed adapter tensors. A full-logit fit failed with GPU OOM before its first update. The successful supervised-suffix-logit implementation retains all context and supervised targets; software equivalence checks and both failed/successful attempts are retained. Teacher-forced loss is not QA accuracy. Training has its own GPU/host memory accounting and is not claimed to fit the CPU inference tier.

The common actor continues from the initial actor with 128 training examples, 32 validation examples and 16 optimizer updates. The matched control critic continues from its recorded initializer with 384 training examples and 48 updates. Its validation groups contain 192 crossed, 192 mutation and 32 replay-critic examples. The continuation recipe uses learning rate 10^{-4}, zero weight decay, clipping norm 1, accumulation 8, one epoch, seed 42, optimizer reset, rank 8, LoRA alpha 16, dropout 0.05 and all-linear target modules. Training versions include PyTorch 2.6.0+cu124, Transformers 4.57.1, PEFT 0.17.1, bitsandbytes 0.48.1 and Accelerate 1.10.1. Initializer, data, optimizer and conversion manifests are retained. Native application checks establish that adapters alter outputs, not that they improve QA. The converted common actor SHA-256 is `4e21beca69dcf7bf2b9e89df7d189c74740d98acebe08501fd7e848bb7155d3e`.

B.4 Cohorts, Exposure and Prompt Availability

FinQA train/development/test snapshots share no exact question or page IDs in the recorded audit but do overlap at company-year level: 243 of 278 test reports appear in the raw training source and 93 in raw development. Exclusion of actually used fitting/validation reports differs from exclusion of an entire public source split. We describe exclusions per study, not as a universal "unseen test" guarantee. The 96-report precision/catalog/format cohort is shared, not

three independent replications. The oracle matrix uses the same 15 enriched reports in every arm. The 72-context TAT-QA format comparison reuses a hash-ordered subset of the 120-context capacity cohort; the separate 72-context adaptation-transfer cohort excludes earlier contexts. The held-out planner primary and secondary cohorts exclude prior table IDs and exact table duplicates, but no report mapping or pretraining audit is available.

The TAT-QA dataset/evaluator source is pinned at repository revision `870accc41953dcde885aabeb963d94aabdc0fbc3`. Downloaded evaluator and data bytes match all recorded hashes. Primary/secondary planner packet hashes are respectively `e463671e33f56af56e6d8d1ad07720b3e75fa3c9a732746a541483fb44251ac0` and `f494a4268baa1623b8159dc2d428c0cfaeb3235e99df5e9831bf923dcaa1a802`, using LF line endings. TAT-QA is attributed to its authors [Zhu et al., 2021]; the data card specifies CC BY 4.0. Third-party source files retain upstream provenance; inclusion does not change their licenses.

Windows resident ledgers retain responses, usage and decisions but do not embed every rendered prompt. Exact prompts are reconstructible from the frozen packet and versioned pipeline/template code. The paper package records these source hashes and includes the relevant template files; it does not falsely describe missing rendered prompts as observed logs. Template text includes question/evidence, source catalog, program grammar and scale instructions; review sees the candidate and deterministic execution without labels. The planner uses only operations, dependencies and VALUE placeholders. Complete reference programs appear only in explicitly labeled oracle diagnostics and controlled candidate challenges.

C Metrics and Statistical Details

Historical reported-answer score. The evaluator parses supported simple numerical reported labels and their rounding precision. Predicted percentage points are divided by 100; percent labels require percent/fraction output units, while ordinary numerical labels require number/fraction units. The normalized value must fall within the larger of reported rounding tolerance and 10^{-5}, with a valid ANSWER decision and no execution issues. Unsupported labels remain in the denominator. This historical rule can disagree with the raw execution reference and should not be interpreted as official program equivalence.

Raw execution score. For finite predicted $\hat{y}$ and raw reference y, the numerical condition is $|\hat{y} - y| \leq \max(10^{-5}, 10^{-5}|y|)$. Correct delivery additionally requires the pipeline to deliver. The stored raw-correct field is a numerical score; source support and review are separate. Primary SAS has 34 deliveries correct by both scores, 44 reported-only, four raw-only and 44 wrong by both. Thus 48/126 delivered answers change correctness label; the two scores are not nested. Primary MAS has five correct by both and three wrong by both.

Official benchmark evaluators. Windows FinQA program outputs are translated to the benchmark's arithmetic token representation and evaluated using the pinned implementation, then compared with stored execution labels. Its evaluator hash is `845cd131cab843eceff256cf6d392978cc470a7da4a80107beb56027fdca5c13`. TAT-QA uses the authors' `TaTQAEmAndF1` implementation, hash `2aeeac479f89f8c76300af1cc0e8d098eb86af84bc386b38b6ab4af484a6dea8`, with predicted numerical scale. On the reported arithmetic single-number comparisons, summed F1 equals summed exact match; it is not an additional independent success endpoint.

Cluster bootstrap. For Study A, let cluster g contain n_g questions and paired difference sum s_g. Each draw samples the 50 reports with replacement and computes $\sum_g s_g / \sum_g n_g$ over sampled clusters. We use 10,000 draws, NumPy `default_rng(42)`, and linear 2.5/97.5 percentiles. The primary reported correct-delivery interval is $[-0.538961, -0.365517]$ and company-cluster sensitivity is $[-0.534722, -0.364405]$. The corresponding raw interval is $[-0.282609, -0.133720]$. These are resampling estimates under the observed cluster structure, not randomized-treatment causal intervals. The original sealed analysis uses Python RNG/quantile indexing; its original interval $[-0.540373, -0.366906]$ is retained, not overwritten.

Planner paired intervals reproduce the archived Python `random.Random(73)` procedure: 10,000 draws of 96 table-context differences, sorted indices 250 and 9749. For gains b, losses c, the two-sided exact McNemar value is $\min[1, 2\sum_{k=0}^{\min(b,c)} \binom{b+c}{k} / 2^{b+c}]$. With four gains and zero losses it is 0.125. Empirical resampling cannot invent a loss absent from four observed discordances, explaining why the positive percentile interval must not be presented as

strong confirmatory evidence. The zero-width secondary correct-delivery interval is a degenerate empirical bootstrap, not proof of population equality.

Older Windows paired summaries generally use 2,000 company-cluster draws with Python seed 42 and indices 50/1949; we retain that procedure instead of silently substituting a different interval. Exact question-pair tests assume independence of pairs; shared companies may violate it. Older 16 GB supplemental analyses include company sensitivity, equal-report weighting, leave-one-report-out checks and cluster sign-flip tests. These post-hoc analyses are restored in Sections J–L; sign-flip assumptions are explicit. The new paper does not elevate them into prespecified primary discoveries. No observed-power, noninferiority or equivalence claim is made. No seed/model uncertainty is inferred from a question bootstrap.

D Complete Main-Study Outcome and Resource Tables

	Reported			Raw			Hours	RSS GB
Profile / arm	C	W	H	C	W	H		
Primary SAS	78	48	34	38	88	34	11.668	10.075
Primary MAS	5	3	152	5	3	152	3.383	10.946
Direct SAS	50	66	44	14	102	44	5.314	10.070
Direct MAS	0	5	155	0	5	155	0.897	10.934
Repair SAS	79	50	31	38	91	31	13.598	10.087
Repair MAS	5	3	152	5	3	152	3.781	10.946

Table 3: All Ubuntu conditions, $n = 160$ each. Hours sum saved inference-stage time, excluding startup, interrupted work and restart overhead. Memory is sampled process-tree RSS in decimal GB.

The primary/repair initial programs, scores, execution and verdicts are matched. The analysis stage intentionally reaches its 128-token allowance on 159 SAS and 157 MAS primary questions; these events are not malformed final JSON or context skips. The sealed analysis reports zero context/budget-skipped calls in the six jobs. Removing unused operations by backward slicing preserves executable numerical values; selecting an earlier endpoint is a different, potentially answer-changing intervention.

Cohort	Arm	C	W	H	Correct candidates
Primary 96	Direct 7B	19	76	1	19
	Program 7B	15	80	1	15
	Current MAS	6	23	67	26
	Planner MAS	10	25	61	25
Secondary 28	Current MAS	2	10	16	5
	Planner MAS	2	10	16	3

Table 4: Held-out TAT-QA official exact-match outcomes. Direct/program candidates equal final predictions because those arms have no review. Secondary 7B jobs are incomplete and are not assigned whole-cohort scores.

Primary baseline outcome	Planner correct	Planner wrong	Planner withheld
Correct	6	0	0
Wrong	2	19	2
Withheld	2	6	59

Table 5: All 96 paired delivery transitions. Four correct gains coexist with six newly wrong deliveries and four removals of wrong delivery (two corrected, two withheld).

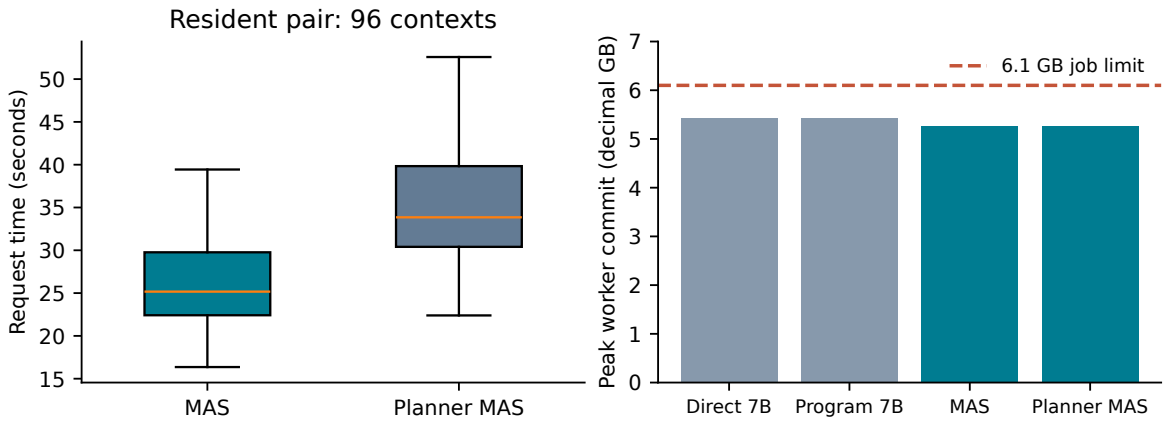


Figure 5: Primary held-out planner resources. Left: distribution of resident request time; boxes show interquartile ranges, median and standard 1.5-IQR whiskers, with outliers omitted from display but retained in data. Right: measured peak worker commit, not physical RAM or an uncertainty interval. Direct/program 7B timing includes load and is therefore reported separately in text.

Secondary transitions are two correct-to-correct, nine wrong-to-wrong, 15 withheld-to-withheld, one wrong-to-withheld and one withheld-to-wrong. Thus equal aggregate wrong counts do not mean identical behavior. Both correct delivery identities are unchanged. Seventeen of 28 predicted shapes are usable.

E Exploratory Sequence: Positive, Null and Adverse Findings

E.1 Earlier Linux Development and Training

The original structured synthetic development suite is solved by a generator-aware deterministic parser (150/150); the 24 temporal KYC probes are also solved by a compact rule control. These are regression/mechanism tests and cannot establish the necessity or superiority of an LLM. A four-case shared-weight pilot improves delivered coverage from zero to one while candidate correctness remains one; patch repair has no gain. Shared residency and successful fitting establish resource behavior, not semantic capability.

On the repeatedly reused first 24 FinQA development questions, the stronger 7B SAS with 128-token analysis and two demonstrations delivers 13 correct, five wrong and withholds six by reported scoring. Longer 256-token analysis yields 11 correct and nine wrong; more table coverage leaves 13 correct. Symbolic cancellation feedback corrects no answers beyond plain retry in a two-trigger pilot. The 3B MAS with analysis/demonstrations produces four reported-correct candidates but delivers three correct, one wrong and withholds 20. Longer analysis does not improve its four correct candidates and increases wrong delivery to ten. These are development comparisons, not optimized-system bounds.

Matched no-analysis/no-demonstration actor adaptation changes 0/24 reported-correct candidates (all refusals) to 11/24, with ten correct, 11 wrong and three withheld. The gain is real on reused data, but starts from a weak floor baseline. A larger unadapted 3B verifier with the same 11 correct drafts withholds all 24, demonstrating that stronger nominal verifier capacity alone does not ensure useful selective delivery.

Matched critic continuation on 52 questions/15 report groups yields control 3/19/30 and crossed 3/15/34 by reported scoring; the wrong-rate interval includes zero. Against the stronger original critic's 4/22/26, crossed loses one correct delivery and fails the no-correct-loss rule. Repair changes no final outcomes relative to saved no-repair prefixes. A reserved 49-question/15-report diagnostic yields control 4/17/28 versus crossed 4/9/36 under reported scoring, but raw correct delivery falls from eight to seven. Selecting only the favorable metric would conceal failure of the both-metric adoption criterion. SAS and additional baseline conditions are all retained in the condition census below.

The matched SAS continuation on the same 52-question selection cohort changes reported outcomes from 9/20/23 (control) to 9/10/33 (crossed), but raw correct delivery decreases from 11 to ten. The unadapted analysis/demonstration baseline gives 9/24/19 reported and 10/23/19 raw. Thus the crossed SAS can improve wrong delivery against that

baseline without an observed correct-count loss, yet still fails the both-metric no-loss comparison against the matched trained control. These are different reference systems, not interchangeable success claims.

On a controlled 12-report/24-program mutation probe, a trained critic accepts 11 of 12 reference programs and zero mutations; the plain reviewer withholds all. A question-blind depth-three program-feature tree scores 275/382 on original synthetic validation, and an extended-corpus tree 295/382 on its own validation; both exceed the 50% majority baseline without seeing questions. On the reused natural24 pool, original/extended trees yield reported 5/8/11 and 5/5/14, versus the original neural critic's 11/4/9. These positive neural and shallow-control results show both discriminative potential and shortcut opportunities, not proof of neural shortcut use or broad semantic understanding. The neural gain remains development evidence.

A source-header materialization prototype adds 183 catalog quantities while preserving all 1,922 original quantities across 73 fitting packets, and restores reference operand coverage on four questions. A proposed 292-case-arm neural comparison was prepared but not run. Executable source support is verified; a QA gain is not measured. It is not counted as a completed method success.

External exploratory controls use original unadapted recipes: MortarBench local typed matching is 39/90 for SAS and 26/90 for MAS, based on partial JSON/XML evidence from nine prepared loan cases; this is not the official full benchmark protocol. The FRAMES closed-book control gives 2/100 and 1/100 under strict matching, unchanged by punctuation sensitivity; evidence URLs are not retrieved. These controls do not establish retrieval or multi-hop generalization. Their scoring and exposure limitations prevent treating them as confirmatory evidence for the adapted financial program systems.

E.2 Windows Candidate and Verifier Pilots

The initial planned 24-pair Windows candidate comparison stops after five complete pairs for post-hoc diagnosis. Refusal-allowed yields five refusals. Forced attempts yield zero reported-correct and one raw-correct candidate; a forced-one-step follow-up on the same five yields two reported-correct and one raw-correct. The original reviewer withholds everything. Removing contradictory repair fields or using a verdict-only 0.5B reviewer accepts all five (two reported-correct, three wrong); a 3B verdict-only reviewer withholds all. This exposes a local schema-policy problem but not semantic discrimination.

On four new reports with reference/near-miss candidates, the 0.5B reviewer accepts all eight and the 3B reviewer withholds all eight. Forced 3B comparison selects the reference in five of eight order presentations, only two of four reports in both orders. On six new reports with a reject-both option, a reversed-order agreement rule yields three correct, zero wrong and nine withheld across the two pair types per report. These dependent, reference-enriched challenges do not estimate natural-pool delivery risk.

The next six-report natural-pair pilot produces no correct final candidate in either draft and withholds all six. A separate six-report base/adapted pilot produces zero versus one correct candidate, yet agreement delivers zero correct, two wrong and withholds four. Thus the apparently error-free constructed result fails to transfer. A six-report adapted-3B versus unadapted-7B program pilot produces one versus zero correct final candidates, with more resources used by 7B; an answer-key prefix selector could recover four reports, but no deployable endpoint selector is demonstrated.

E.3 Endpoint, Retrieval, Precision and Catalog Controls

Model-predicted operation count on 16 reports changes official correct candidates from one to three (three gains, one loss). A gold-count control also yields three, on exactly the same recovered reports; gold count adds no recovery beyond predicted count on that slice. A retrospective endpoint selector makes no change in its initial test. On the 96-report matched precision cohort, a first-step rule changes Q4 official accuracy from 2 to 8 and Q8 from 2 to 6. On the earlier 24 reports it instead reduces Q4 from 3 to 1 and Q8 from 5 to 1. This cross-cohort reversal prevents a general recommendation to truncate at the first operation. The answer-key prefix oracle reaches at most 11/96 per arm, leaving most errors outside prefix selection.

Unused operations occur in 67/96 Q4 versus 14/96 Q8 outputs (and 20/24 versus 7/24 in the earlier cohort). Slicing preserves final execution results; it does not create an accuracy gain. Precision raises initial official matching from 3/24 to 5/24 (one loss, three gains; $p = 0.625$), with unchanged 3/24 normalized symbolic matches. Its 96-report replication is 2 versus 2; reported scoring also ties but with one discordance in each direction rather than the two official discordances each way. Q8 raises peak commit from about 2.59 to 3.94 GB and median latency from 20.6 to 32.2 seconds in that replication.

Retrieval screens on 883 development questions cover all annotated spans for lexical8 750, BM25 748, reciprocal-rank fusion 733, BGE-small 728 and BGE/lexical hybrid 748. Wider lexical retrieval covers 827, adding 77 with no losses, at a larger token allowance (median evidence/catalog tokens 711 to 1,225). Numerical reference bindability rises 703 to 767; source identity and model success remain separate. The pilot's 1/16 to 4/16 generation gain ($p = 0.25$) fails directionally at 2/21 to 1/21 ($p = 1$). The latter company-cluster interval is $[-20.8, +11.1]$ percentage points, and median latency rises 18.9 to 26.3 seconds. A fixed-catalog follow-up was proposed to separate additional text from catalog options, but no completed result is identified in the retained retrieval evidence; it is not counted as an executed ablation.

Catalog completion makes 28 additional annotated test programs numerically bindable in its source screen. Five enriched neural cases improve 0 to 2; on the unselected shared 96 reports accuracy changes 2 to 1, despite reference-executable witnesses increasing 82 to 84. On the separate 15-report oracle matrix, repeated unrestricted controls reproduce their plans. All four structure-only successes remain correct with the shortlist. Structure plus shortlist yields ten official and seven reported successes; it fixes multiple variables simultaneously and is neither a deployable method nor a single-factor mechanism attribution.

E.4 Baseline Capacity, Format, Review and Reference Checks

The initial 16-context TAT-QA format pilot gives 1/16 for both 3B direct and program output, on different questions. The 40-context direct-capacity comparison gives 5/40 for 3B and 6/40 for 7B (four 3B-only, five 7B-only); a 120-context replication gives 12 versus 29 (five versus 22 unique successes). Gold-scale substitution rescues 11 wrong 3B and five wrong 7B answers in that replication, showing that scale selection matters but does not explain the full gap. This is an answer-key sensitivity, not deployed accuracy.

Fixed-7B format controls favor direct output on FinQA's shared reported metric, 35/96 versus 20/96, but favor programs with substantial uncertainty on TAT-QA, 19/72 versus 14/72. The FinQA agreement gate retains only three correct, two wrong and withholds 91, losing 17 of 20 reported-correct programs. Its percentage-label subgroup is direct 23/53 versus program 19/53; other supported numerical labels are 12/40 versus 1/40; three unsupported labels receive no match. The original official program score 1/96, converted sensitivity 19/96 and reported score 20/96 are retained separately.

The initial trained-critic replay on 16 wider-evidence candidates appears favorable under reported scoring: control keeps three correct with zero wrong. The transfer replay on 21 narrow candidates has an executable-only baseline of two correct/19 wrong; control yields 1/5/15, crossed 1/2/18. Both lose one correct candidate, and two of three crossed-accepted outputs are wrong. Independent 7B recomputation on the initial 16 yields four correct answers alone, but the agreement gate withholds all; one overlapping correct answer pair differs beyond the frozen tolerance despite matching the reported label separately.

On 72 separate TAT-QA contexts, actor adaptation improves candidates from 3 to 8, and review delivers 6/10/56; ten of 16 accepted answers are wrong. A resident 21-report workload produces six official-correct (five reported-correct) candidates and delivers 5/4/12 under both scores. Disabling RAM prompt caching preserves all plans/verdicts/deliveries, reduces sampled RSS from 5.852 to 5.013 GB and job peak from 6.101 to 5.264 GB. The default-cache job logs failed allocation and fallback, whereas the disabled-cache run logs none. A failed-allocation peak does not identify exactly how many bytes were successfully committed.

A temporal growth rule applied to 453 questions from 117 report-disjoint test reports delivers only three, none matching either benchmark score. Individual discrepancies include plausible financial interpretations and incompatible reference conventions; they are not adjudicated gold-label corrections. A broader consistency screen reproduces every stored annotated execution result for 883 development and 1,147 test questions, then flags reported/raw incompatibility on 110/851 eligible development labels and 129/1,100 eligible test labels. These are screening flags, not 239 proven annotation errors. Official primary labels and all cases are retained.

E.5 Planner Development and Held-Out Result

The initial 12-report prototype produces a usable shape on only one report; 11 predictions are constant-only. Revising the grammar to unbound VALUE slots yields usable shapes on 11/12. Official final correct deliveries remain three in both arms; wrong decreases four to three and withheld increases five to six. Reported-correct delivery falls five to four. Median latency rises 23.0 to 30.9 seconds. This is development evidence for producing valid shapes, not an accuracy gain. The revision precedes held-out inference.

In the main 96-context cohort the four correct-delivery gains have no losses, but wrong delivery rises two and candidate correctness falls one. The main paper reports this failed practical criterion prominently. At 28 secondary contexts there is no correct-delivery change, and two correct candidates are lost. The incomplete 7B test prefix and recovery attempt remain archived; completion of MAS does not imply completion of the four-arm comparison. No new test experiments are substituted for those failures.

F Resource Failures, Missing Evidence and Scope of Claims

Early Linux training OOMs, converter/application interface issues, capacity probes, historical process swap, interrupted external evaluation, default-cache allocation fallback, secondary 7B allocation failures and Ubuntu SSD disconnections are retained. They are neither silently excluded nor pooled into successful end-to-end timing. The original 16 GB recovery run's score completeness did not validate its no-swap claim. The Ubuntu run supplies a separate resource-qualified repeat, with the same questions and scores.

The SSD was mounted read-only after NTFS metadata errors. Six unreadable replay JSONLs were reconstructed using sealed analysis code and pinned 0.3.16 vocabulary-only tokenizers; their hashes, all 40 analysis-evidence hashes and the entire summary match the original audit. The text report was regenerated without an original recorded hash. The original results tarball and one bootstrap log remain unavailable, and a workflow marker still reads RUNNING. Therefore evaluation and replay completion are supported, but successful original packaging is not. Repository handoff ZIPs verify independently. Model tensors were not rehashed during manuscript preparation and are not included in portable evidence.

Missing or limited evidence includes physical 8 GB/no-paging validation, an uninterrupted full Ubuntu run, complete secondary 7B baselines, TAT-QA report-level grouping, multiple model families/seeds, independent financial annotation adjudication, optimized baselines for every method and matched-latency system initialization. None is filled with estimated or fabricated data. No measured 32 GB result exists. The paper makes no calibrated safety, financial compliance, causal architecture or universally optimal precision claim.

G Literature Verification and Reproduction Commands

The original focused bibliography contains 17 references verified against primary arXiv metadata, ACL Anthology or PMLR publisher BibTeX. The accompanying literature audit records the claim supported by each source and comparison limits. Recent financial verification papers are identified as arXiv preprints; we do not invent venues or reproduce their performance under our budgets. Our numerical results come only from this repository, not from unrelated published leaderboards. LoRA, QLoRA, program execution, selective prediction and multiagent debate are prior methods, not originality claims.

From a repository checkout, restore immutable evidence into a separate directory, substituting its path for PATH:

```
python dashboard_handoff/restore_evidence.py --extract PATH
python scripts/reporting/build_final_paper.py --evidence PATH
python scripts/reporting/verify_final_scores.py --evidence PATH
```

These commands use preserved predictions and do not load model weights or run inference. Dependencies and exact input hashes are recorded with the paper. Compile each LaTeX source using PDFLaTeX, BibTeX and two additional PDFLaTeX passes. The main paper uses unmodified AAAI-2027 style/BibTeX files; this separate technical supplement uses one column for audit readability. Submission-specific administrative declarations, final author identities and any required checklist require author review; none is fabricated.

H Historical Linux 8 GB Condition Census

The following is generated directly from the preserved condition inventory, without selecting conditions by outcome. It covers the exported natural-task development/selection/diagnostic conditions. Separate Windows experiments, synthetic regression tasks, external controls and unsuccessful execution attempts are described above and retained in the machine-readable artifact inventory. The labels retain original run names to make each row findable. R and X are the disclosed reported and raw historical scores, not interchangeable with official FinQA equivalence.

Table 6: All 15 earlier FinQA conditions, including sampled memory. Cohorts overlap; these are not 15 independent replications. Reported and raw C/W/H retain every case.

Condition	n	Reported C/W/H	Raw C/W/H	RSS GB
diagnostic_audit-control_mas-full-01	49	4/17/28	8/13/28	5.052
diagnostic_audit-control_sas-full-01	49	8/20/21	14/14/21	5.368
diagnostic_audit-crossed_mas-full-01	49	4/9/36	7/6/36	5.064
diagnostic_audit-crossed_sas-full-01	49	9/20/20	15/14/20	5.364
model_selection-baseline_mas-full-01	52	2/19/31	3/18/31	5.058
model_selection-baseline_original_classic_mas-full-01	52	4/18/30	6/16/30	5.046
model_selection-baseline_original_critic_common_actor-full-01	52	4/22/26	6/20/26	5.056
model_selection-baseline_sas-analysis128-demos2-full-01	52	9/24/19	10/23/19	5.368
model_selection-baseline_sas-full-01	52	8/25/19	10/23/19	5.348
model_selection-control_mas-full-01	52	3/19/30	5/17/30	5.060
model_selection-control_sas-analysis128-demos2-full-01	52	9/20/23	11/18/23	5.365
model_selection-control_sas-full-01	52	8/23/21	10/21/21	5.356
model_selection-crossed_mas-full-01	52	3/15/34	5/13/34	5.067
model_selection-crossed_sas-analysis128-demos2-full-01	52	9/10/33	10/9/33	5.370
model_selection-crossed_sas-full-01	52	10/16/26	12/14/26	5.368

I Original Questions, Design Rationale and Historical Controls

I.1 Questions and hypothesis dispositions

The original four questions ask about correct/wrong/withheld delivery under common ceilings (RQ1), the stage at which failure appears (RQ2), the analysis-plus-demonstration bundle and repair (RQ3), and memory, latency and time per correct delivery (RQ4). They are retained alongside the later question about translating reference-informed diagnostic potential into evidence-only planning. They are related questions, not one pooled causal experiment.

H1 is the primary operational proposal that the resident F16 team reduces wrong delivery without losing correct delivery. Its observed failure is substantial: reported correct delivery falls 78 to 5 while wrong delivery falls 48 to 3. H2 concerns the bundled prompt change: reported correct delivery increases 50 to 78 for SAS and 0 to 5 for MAS. That is descriptive support, not identification of the effect of analysis alone or demonstrations alone. H3 requires repair to increase correct delivery without increasing wrong delivery; SAS adds one correct and two wrong, while MAS changes neither. H4 predicts no higher team resident memory from its smaller parameter count; the observed peaks refute that prediction for these recipes. These are operational dispositions, not four independent discoveries. The frozen primary recipe and outcome definitions precede inference; the expanded significance-test battery, strata, loss functions and mathematical exposition were added later.

The original gap concerning common resource limits is addressed only for the measured recipes and control mechanisms. The gap concerning fair baselines remains partly open: shared packets, tools and ceilings remove some differences, but not model size, precision, role allocation, training or prompts. The adaptation-versus-verification question has exploratory evidence, including correct-answer losses. The proposed compliance-metric gap is not closed by numerical QA. Both synthetic systems miss every planted mismatch and required missing-field refusal. No professional financial or legal validity follows.

I.2 Fairness, memory accounting and protocol choices

The original budget rule was $M_{\text{job}} = \min(0.8M_{\text{tier}}, M_{\text{host}} - M_{\text{idle}} - 1\,\text{GB})$, with all GB decimal. The measured 16 GB host exposes approximately 15.44 GB to Linux. A 12.49 GB-cap attempt failed its reserve check; the frozen evaluation uses an 11.5 GB application RSS guard plus 1 GB reserve and 0.1 GB launch cushion (12.6 GB available before launch). This is a sampling/launch policy, not an OS guarantee that all applications together never exceed memory.

The original 8,192-context engineering probes swapped. Reducing both arms to 4,096 before the accuracy run gave neutral-input sampled peaks of approximately 9.98 GB for SAS and 10.85 GB for MAS without observed swap in those probes. The comparison keeps eight CPU threads, no GPU layers, no memory mapping, greedy decoding, seed 42, F16 KV cache and reset-per-call policy. Sharing the same total generation ceiling does not imply equal actual generated or prompt tokens, model calls, elapsed time, FLOPs or energy. FLOPs and energy were not measured; an older sentence implying otherwise is corrected here.

Study A uses instruction-tuned checkpoints without project adapters. Earlier adapters target 7B/3B bases and cannot simply be applied to 14B. Applying an existing adapter on only one side would introduce another asymmetry, but the absence of such adaptation does not make either recipe optimal. The 8 GB adapted experiments, historical synthetic team, and 16 GB F16 team use different systems. Separate model residency is not shared-backbone adapter serving, and a single model can make several stage calls.

Study A role	GGUF precision	File bytes
SAS 14B	Q4_K_M	8,988,110,336
MAS 1.5B extractor	F16	3,093,669,024
MAS 3B drafter	F16	6,178,316,832
MAS 0.5B reviewer	F16	994,156,512
MAS total	Separate files	10,266,142,368

Table 7: Recorded model-file sizes, distinct from RSS, commit, KV-cache size and total host memory.

I.3 Historical synthetic test outcomes, including denominator failures

The original Windows study tests 100 synthetic KYC and 75 synthetic credit cases, using a 7B Q4 SAS and a resident 1.5B/0.5B/0.5B F16 MAS at context 2,048 and a 6.4 GB sampled cap. It is distinct from the later 3B quantized resident pair. The 150-case development suite is rule-solvable; these tasks provide output-contract and pipeline diagnostics, not evidence that language models are necessary or that the system performs real regulatory work.

Metric	SAS	MAS
Valid JSON / all test cases	175/175	111/175
Invention flag / complete claim sets	125/175	89/111
Grounded claims / evaluated claims	541/641	233/641
Detected planted mismatch / eligible	0/20	0/20
Correct missing-field refusal / eligible	0/59	0/59
Generated tokens / case (mean)	327.5	678.4
Inference seconds / case (mean)	91.5	60.6
Peak recorded RSS (GB)	5.030	5.540

Table 8: Historical 175-case outcomes re-aggregated from saved case scores. The invention denominators differ because 64 MAS outputs are malformed; 80.2% versus 71.4% is not a matched unconditional error comparison. Grounding is claim-level, not an independent sample of 641 tasks.

Credit-only MAS validity is 52% and claim grounding 18.2% in the original summaries. Development hardening changed valid JSON from 100% to 19.3% for SAS and 62% to zero for MAS; the recorded workflow reverted to the original baselines before opening the test set. The original closed-book FRAMES check was SAS 1/100 versus MAS 2/100; the later external control reported above reverses that ordering at 2/100 versus 1/100. These are different runs, not an inconsistency to erase or evidence of retrieval competence.

The historical Windows memory-mapped configuration reported approximately 6.59 GB working-set RSS at context 4,096 and 6.47 GB at 2,048 for a 4.68 GB model; direct loading avoided that observed cap breach. This does not establish that all Windows mmap accounting is erroneous. Combining Q4_0 KV cache and flash attention on that CPU produced zero valid JSON out of 150 development cases; F16 KV without flash attention was used thereafter. The two settings were changed together, so the cause is not separately identified. A later shared-backbone probe reduced sampled RSS from approximately 4.83 to 2.57 GB with matching probe logits; it did not establish end-to-end financial accuracy or substitute for the separately resident experimental systems.

J Extended Study A Outcomes and Resource Reconciliation

J.1 Paired transitions and selective behavior

Under reported scoring the team has no uniquely correct delivery: the paired correct gains/losses are 0/73. Under raw scoring the gains/losses are 1/34. Therefore the unqualified old statement that MAS never succeeds where SAS fails was too broad. Reported wrong gains/losses are 1/46; raw wrong gains/losses 0/85. Withholding gains/losses are 118/0 under both metrics. The reported transition table decomposes its Bowker statistic as $1 + 72 + 46 = 119$; the raw table gives $1 + 34 + 84 = 119$. The dominant movement is delivery to withholding, not correction.

Primary reported precision among delivered answers is 78/126 for SAS and 5/8 for MAS. Case-IID Wilson 95% intervals are approximately [53.2,69.9]% and [30.6,86.3]%, respectively; they are descriptive because selected deliveries differ and questions share reports. Primary correct-delivery Wilson intervals are [41.1,56.4]% and [1.3,7.1]%; wrong-delivery intervals are [23.4,37.5]% and [0.6,5.4]%. These cannot replace paired cluster intervals or justify equal reliability.

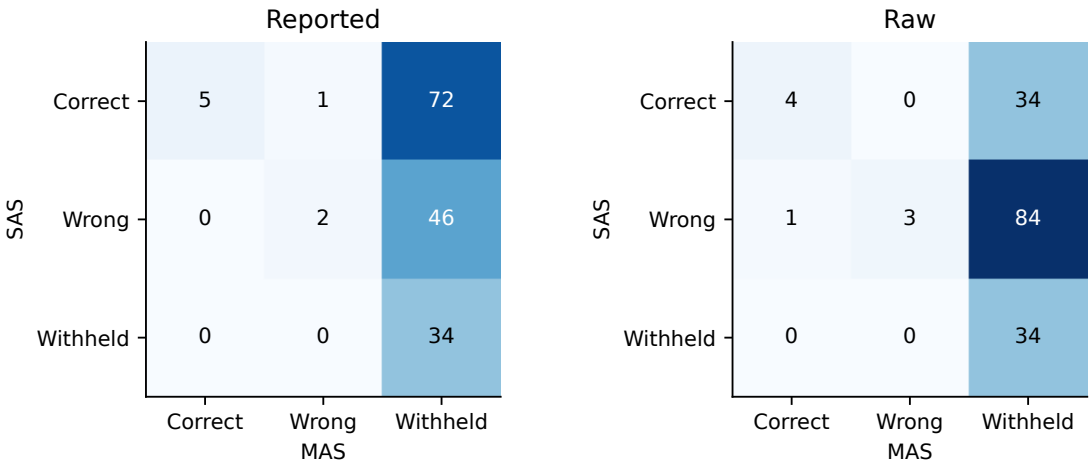


Figure 6: Every primary question in the paired C/W/H transition tables. The numerical score changes which deliveries count as correct but never changes delivery itself. These are observations on the same 160 questions, not separate cohorts.

J.2 Four endpoints, weighting and influence

Table 9: Four Study A endpoints: MAS minus SAS, percentage points. Intervals use 50 report clusters; company sensitivity uses 36. The four 98.75% marginal intervals target approximate family coverage, not an exact finite-sample guarantee.

Endpoint	Δ	Report 95%	Report 98.75%	Company 95%
Reported correct	-45.62	[-53.90, -36.55]	[-56.25, -34.09]	[-53.47, -36.44]
Reported wrong	-28.12	[-37.09, -20.13]	[-39.34, -18.08]	[-38.36, -19.76]
Raw correct	-20.62	[-28.26, -13.37]	[-30.29, -11.76]	[-28.15, -14.07]
Raw wrong	-53.12	[-61.90, -44.08]	[-64.49, -41.26]	[-61.99, -43.08]

Table 10: Weighting and influence sensitivity. Equal-report weighting estimates a different target. Deletion ranges are not confidence intervals; sign-flip tests require null sign symmetry.

Endpoint	Equal-report Δ	95% interval	Leave-one-report range	Sign-flip p
Reported correct	-42.48	[-52.66, -32.31]	[-46.79, -44.23]	1.53e-05
Reported wrong	-30.62	[-40.75, -20.71]	[-29.53, -26.75]	1.53e-05
Raw correct	-18.74	[-26.17, -11.83]	[-22.15, -19.08]	1.53e-05
Raw wrong	-54.37	[-64.00, -44.50]	[-54.61, -51.92]	1.53e-05

Table 11: Primary supplemental cluster diagnostics. CR1 uses normal tails; score uses chi-square(1). Right/left/tied counts refer to report-level event-rate differences, not independent confirmations.

Endpoint	CR1 SE pp	DEFF	CR1 p	Report score p	Company score p	R/L/tied
reported C	4.45	1.28	1.19e-24	6.41e-07	0.000127	0/33/17
reported W	4.40	1.45	1.71e-10	1.84e-06	3.18e-05	1/29/20
raw C	3.80	1.31	5.83e-08	4.26e-05	0.000112	0/21/29
raw W	4.57	1.34	2.96e-31	6.36e-08	5.81e-05	0/41/9

Holm-adjusted report sign-flip values for the four correct/wrong endpoints are all 6.1034×10^{-5}. The corresponding report score-test adjusted values are 1.92×10^{-6}, 3.68×10^{-6}, 4.26×10^{-5} and 2.54×10^{-7} in reported-C, reported-W, raw-C, raw-W order. These are post-hoc diagnostics under different assumptions, not multiple independent confirmations. Three sign-flip results have zero more-extreme Monte Carlo draws, and one has one; their values reflect the finite 65,536-draw resolution. Company score-test p-values are larger (for reported correct, approximately 1.27×10^{-4}). The old claim that every cluster-aware adjusted value is at most 6.1×10^{-5} does not extend to company-level variants.

On reported correct delivery SAS is higher on 33 reports, ties on 17, and is lower on none; raw correct delivery has 21 reports favoring SAS and 29 ties despite one question-level MAS gain. The question-weighted and equal-report estimands differ. Leave-one-report-out ranges show observed influence, not sampling confidence. Deterministic hash-based selection and potential common-company dependence limit population interpretation.

J.3 Candidate production, reference strata and scoring discordance

Arm	Executable	Correct draft	C	W	Correct rejected	Wrong rejected	Invalid
SAS	150	84	78	48	6	18	10
MAS	10	5	5	3	0	2	150

Table 12: Primary reported-score candidate/gate accounting. Invalid includes INSUFFICIENT. A review PASS does not override invalid execution.

Writing q for executable fraction, p for correct given executable, a for acceptance of correct candidates and b for acceptance of wrong executable candidates gives SAS $(q, p, a, b) = (150/160, 84/150, 78/84, 48/66)$ and MAS $(10/160, 5/10, 5/5, 3/5)$. Thus the reviewer accepts 72.7% and 60% of wrong executable candidates, respectively. The equality $C/N = qpa$ is exact descriptive accounting; it does not identify which intervention would change any factor independently. In the later planner study the candidate pool changes, so this fixed-pool calculation must be repeated for each arm.

Table 13: Study A primary reported-score outcomes by reference program (post hoc). Step and operation-family partitions overlap one another; do not sum across partitions.

Reference stratum	n	SAS C/W/H	MAS C/W/H
1 step	104	52 / 29 / 23	5 / 3 / 96
2 steps	49	26 / 14 / 9	0 / 0 / 49
3+ steps	7	0 / 5 / 2	0 / 0 / 7
single add/subtract	40	16 / 12 / 12	1 / 0 / 39
single ratio/product	59	34 / 14 / 11	4 / 3 / 52
multi-step arithmetic	56	26 / 19 / 11	0 / 0 / 56
table aggregate	5	2 / 3 / 0	0 / 0 / 5

There are 104 one-operation, 49 two-operation, four three-operation and three five-operation reference programs. All 56 multi-operation questions are withheld by MAS. Five questions use a table aggregate; the source-reference grammar does not directly implement max/min/sum/average. A reference operation is not necessarily the only valid solution, so grammar mismatch is a threat to comparability, not proof of impossibility.

All annotated spans are retrieved on 138 questions. SAS delivers 74 reported-correct with three invalid/insufficient programs there; MAS delivers five with 128 invalid/insufficient. In the remaining 22, SAS has four correct and MAS zero. Annotated retrieval completeness is a reference-dependent diagnostic, not proof that the question is unambiguous, that all cell identities are bound correctly, or that prompting makes the evidence usable.

Among 126 SAS deliveries, reported/raw agreement is 34 correct under both, 44 reported-only, four raw-only and 44 incorrect under both. All eight MAS deliveries agree across scores. A supported answer can therefore fail one convention while passing another. Neither convention is silently substituted for official FinQA execution scoring, and unsupported labels remain in denominators.

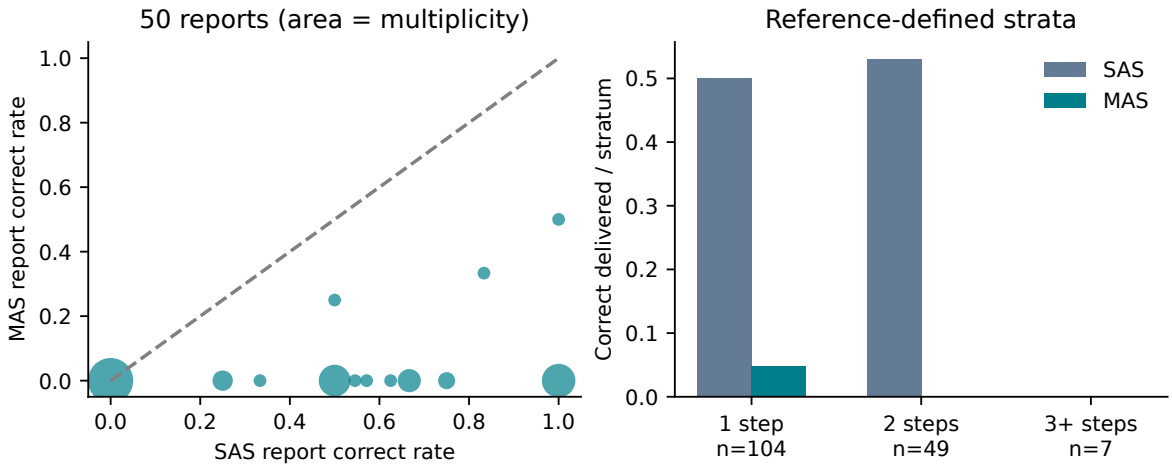


Figure 7: Report heterogeneity and reference-defined complexity. Repeated report-rate pairs are drawn with proportional marker area. Right: observed correct-delivery rates; stratum counts and all wrong/withheld outcomes are in Table 13. These exploratory partitions do not isolate causal difficulty.

J.4 Prompt, repair and resource costs

The full paired battery below retains the direct-to-primary and primary-to-repair contrasts under both scores. SAS reported bundle gain is 28/160 with report interval [+9.76,+24.55] pp; its raw gain is 24/160 with [+8.59,+21.62]. MAS gains 5/160 with [0,+6.98]. The SAS raw wrong-delivery decrease has interval $[-17.96, +0.72]$ pp: it should not be called robust merely because the case-IID exact test is near 0.05. The two prompt components are inseparable here.

Primary and repair initial candidates match on all cases. SAS repair releases one reported-correct and two wrong answers (zero raw-correct, three raw-wrong); MAS attempts 152 repairs without changing delivery. Final source-supported MAS candidates fall from ten to eight in the original trace accounting. A degenerate no-change interval does not prove equivalence in future cases.

Table 14: Timing and sampled-RSS reconciliation across original recovery and Ubuntu repeat. Summed inference excludes aborted work, model loading and restart overhead. Seconds per correct is amortized measured work, not expected retries to solve a question; undefined at zero correct.

Run	Recipe	Hours	Median s	s/correct	Output tok	RSS GB
Historical	mas-direct control	0.821	17.5	–	8392	10.901
Historical	mas-primary	2.432	53.8	1751.4	29509	10.929
Historical	mas-repair ablation	3.682	81.9	2651.2	33917	10.913
Historical	sas-direct control	5.266	112.2	379.1	14449	10.040
Historical	sas-primary	11.546	251.4	532.9	32657	10.062
Historical	sas-repair ablation	13.494	259.9	614.9	35244	10.074
Ubuntu	mas-direct control	0.897	18.5	–	8392	10.934
Ubuntu	mas-primary	3.383	54.6	2435.7	29509	10.946
Ubuntu	mas-repair ablation	3.781	83.7	2722.6	33917	10.946
Ubuntu	sas-direct control	5.314	112.7	382.6	14449	10.070
Ubuntu	sas-primary	11.668	252.4	538.5	32657	10.075
Ubuntu	sas-repair ablation	13.598	260.4	619.7	35244	10.087

Table 15: Primary stage totals in both runs. Prompt tokens count every call; they are not one context window. Stages execute sequentially while model weights remain resident.

Run	Arm	Stage	Hours	Output tok	Prompt tok
Hist.	MAS	extractor/select_evidence	0.237	4121	128047
Hist.	MAS	drafter/analyze_question	0.960	20441	134127
Hist.	MAS	drafter/program	1.186	3507	390092
Hist.	MAS	verifier/verify_program	0.049	1440	94407
Hist.	SAS	sas/analyze_question	2.952	20479	134127
Hist.	SAS	sas/program	6.656	11058	383588
Hist.	SAS	sas/verify_program	1.938	1120	124971
Ubuntu	MAS	extractor/select_evidence	0.346	4121	128047
Ubuntu	MAS	drafter/analyze_question	1.264	20441	134127
Ubuntu	MAS	drafter/program	1.701	3507	390092
Ubuntu	MAS	verifier/verify_program	0.072	1440	94407
Ubuntu	SAS	sas/analyze_question	2.983	20479	134127
Ubuntu	SAS	sas/program	6.725	11058	383588
Ubuntu	SAS	sas/verify_program	1.961	1120	124971

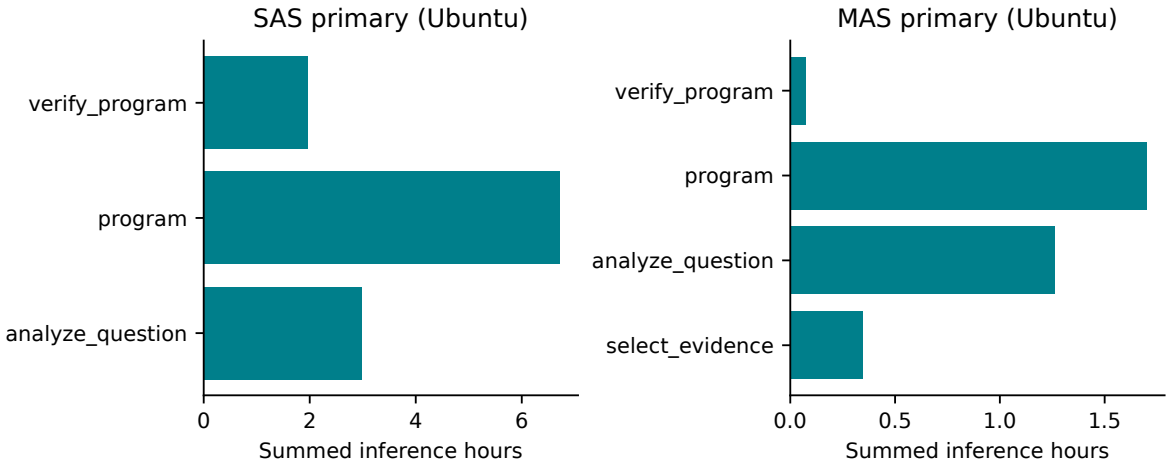


Figure 8: Stage work in the Ubuntu primary pair, summed from saved inference durations. This exposes coordination work but does not isolate a causal overhead penalty: models and prompts differ. Startup, aborted work and recovery time are excluded.

Original primary medians are SAS 251.4 s and MAS 53.8 s; original amortized seconds per reported-correct answer are 532.9 and 1,751.4. Hence the historical statement of approximately 3.29 times more work per correct answer applies to that run only. Original primary prompt tokens average 4,016.8 and 4,666.7, while generated tokens average 204.1 and 184.4. The Ubuntu repeat has different durations even though scores agree. It must not inherit the original speed ratios or the original MAS repair increment of 1.25 hours. Table 14 reports both runs distinctly.

The historical chronology is: a failed reserve check; context reduction after swap-prone probes; SAS primary swap of 44,011,520 bytes after 145 saved cases; a replay-checked continuation that later records 34,430,976 bytes of swap after seven direct-control cases; and a final continuation preserving 327 cases under kernel swap prevention. The original complete accuracy audit remains resource-invalid as an uninterrupted no-swap test. The separate Ubuntu repeat contributes zero observed swap over its saved samples, not a retroactive erasure of those failures. Its own interruptions and damaged export are disclosed in the resource section above.

K Conditional Mathematical Analysis and Statistical Assumptions

The following are established identities and conditional bounds, not novel theorems. Counts are capital C, W, H; rates divide by N. Probabilistic claims require the stated sampling assumptions and do not follow solely from deterministic trace replay.

K.1 P1–P3: accounting, the gate bound and stage factorization

For each case exactly one of correct delivery, wrong delivery and withholding occurs, hence $C + W + H = N$ and $\Delta C + \Delta W + \Delta H = 0$. Proof: sum the exhaustive, mutually exclusive indicators. Always-withhold gives zero errors but also zero correct delivery. This identity explains why fewer wrong answers alone does not establish improvement.

For a fixed candidate set, let A be the cases with correct executable candidates and D the delivered cases. A gate that only accepts or rejects obeys $C_{\text{gate}} = |A \cap D| \leq |A|$. Proof: an intersection is a subset of A. This bound assumes no rewrite, repair, new evidence or alternative candidate selection. Study A MAS has $|A| = 5$, so an ideal gate over its existing drafts cannot reach SAS's 78 reported-correct deliveries. It does not rule out improved generation, representation, role allocation or a different candidate pool.

Let E denote an executable candidate and K numerical correctness, with delivery requiring E. Define $q = \Pr(E)$, $p = \Pr(K \mid E)$, $a = \Pr(D \mid E, K)$ and $b = \Pr(D \mid E, \neg K)$. Then

$$C/N = qpa, \qquad W/N = q(1-p)b, \qquad H/N = 1 - qpa - q(1-p)b. \tag{4}$$

These empirical-frequency equalities follow from the chain rule, not independence. When a conditioning set is empty its conditional rate is undefined; the corresponding joint count remains zero. Reporting fictitious zero conditional accuracy in such a cell would be misleading.

K.2 P4–P5: preference-sensitive utility and corrected repair break-even

For chosen penalties $\lambda, \kappa \geq 0$, define

$$U = (C - \lambda W - \kappa H)/N. \tag{5}$$

The original observed primary reported-score difference is

$$U_{\text{SAS}} - U_{\text{MAS}} = (73 - 45\lambda + 118\kappa)/160, \tag{6}$$

and the raw-score difference is $(33 - 85\lambda + 118\kappa)/160$. Proof: substitute the observed counts. Thus switch points are $(73 + 118\kappa)/45$ and $(33 + 118\kappa)/85$, respectively. These are sample-specific preference calculations, not calibrated institutional risk thresholds or prospective optimization results.

At $\kappa = 0$, reported-score MAS beats SAS only for $\lambda > 73/45 \approx 1.622$ and beats always-withhold only for $\lambda < 5/3 \approx 1.667$. Both constraints are necessary for MAS to be best among these three policies. With withholding cost, the MAS-versus-always-withhold condition becomes $\lambda < (5 + 8\kappa)/3$. The raw-score SAS/MAS switch at $\kappa = 0$ is $33/85 \approx 0.388$; one must not transfer the narrow reported-score optimality region to raw scoring. No empirical estimate of the appropriate penalties is available.

Repair changes SAS by $(\Delta C, \Delta W, \Delta H) = (1, 2, -3)$ reported and $(0, 3, -3)$ raw. Consequently

$$\Delta U_{\text{rep}} = (1 - 2\lambda + 3\kappa)/160, \qquad \Delta U_{\text{raw}} = (-3\lambda + 3\kappa)/160. \tag{7}$$

The old manuscript omitted 3κ; its formula is correct only at zero withholding cost. Subtract an additional measured-cost term $\gamma \Delta t$ if time is priced per question. A positive time cost lowers repair utility, but does not automatically make it negative for every penalty choice. MAS's unchanged outcomes give zero outcome-utility change; additional work is unfavorable if time has positive cost. Under the prespecified correct/wrong criterion SAS repair fails regardless of a post-hoc favorable loss function.

K.3 P6: memory accounting with explicit residency assumptions

If each distinct tensor j has n_j stored elements with effective storage b_j bits per element, its storage is $n_j b_j / 8$. If all such storage is actually resident simultaneously without swapping/compression, and runtime buffers occupy disjoint

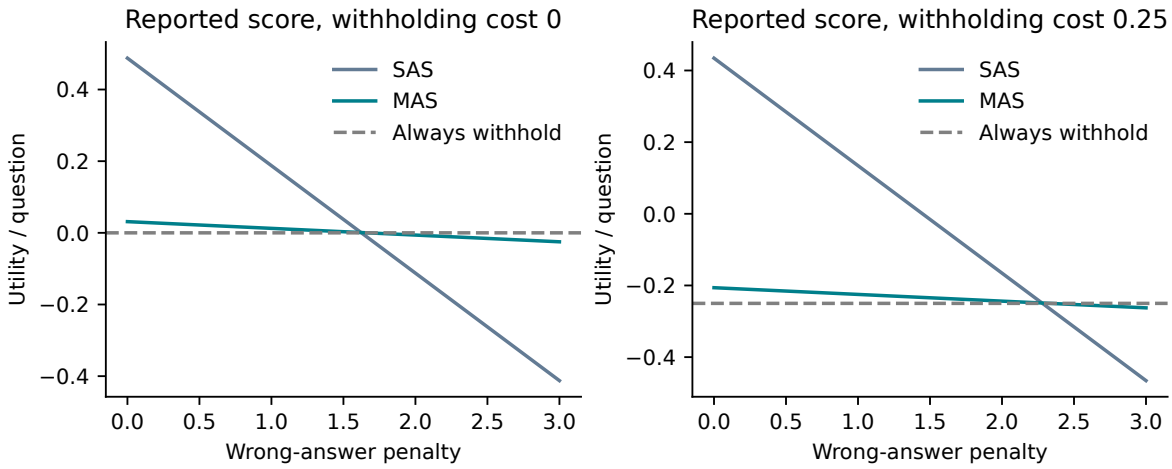


Figure 9: Observed reported-score utility with the always-withhold policy included. Curves are exact transformations of counts, not fitted predictions or confidence bands. Loss weights were not estimated from a financial institution.

resident pages, a bookkeeping bound is

$$M_{\text{resident}} \geq \sum_j n_j b_j/8 + M_{\text{KV}} + M_{\text{other}}. \tag{8}$$

Proof: add the disjoint resident allocations, counting a truly shared tensor once. Effective bits must include block scales and other encoding overhead. File headers, padding, shared mapped pages and allocator reporting prevent treating file size as an exact RSS component. Summed process RSS may double-count shared pages; the experiments use no mmap, but this is not a universal memory-accounting theorem. Windows commit is a different measure. Table 7 explains why fewer parameters can occupy more storage at higher precision without claiming a causal precision effect on accuracy.

K.4 P7: bounded-cluster uncertainty, with its limits

For independent report contributions $D_g \in [-1, 1]$ and fixed weights $w_g = n_g/N$, Hoeffding's inequality [Hoeffding, 1963] yields

$$\Pr\left(\left|\sum_g w_g(D_g - \mathbb{E}D_g)\right| \geq \epsilon\right) \leq 2\exp\left(-\frac{\epsilon^2}{2\sum_g w_g^2}\right), \qquad G_{\text{eff}} = 1/\sum_g w_g^2. \tag{9}$$

The range of $w_g D_g$ is $2w_g$; substituting these ranges into the bounded-sum inequality proves the expression. Setting its right side to α gives half-width $\sqrt{2\sum_g w_g^2 \log(2/\alpha)}$. Replacing α by $\alpha/4$ and applying the union bound gives a four-endpoint bound, assuming all four marginal conditions hold.

Observed sizes give $G_{\text{eff}} = 36.99$ for reports and 19.88 for companies, with 95% half-widths 44.66 and 60.93 pp. The report four-endpoint half-width is 52.38 pp. The reported-correct -45.625 pp difference barely excludes zero with the report single-endpoint bound, but not with the family or company bound. This is conservative conditional uncertainty for fixed observed weights, not a design-based guarantee for deterministic report selection, a random-denominator population ratio, or dependence across companies. The effective-cluster number is weight concentration, not the literal number of independent reports.

K.5 P8–P10: score disagreements and procedure-specific evidence bounds

P8 follows by partitioning deliveries into both-correct, reported-only, raw-only and neither: $C_{\text{rep}} - C_{\text{raw}} = R_{\text{only}} - E_{\text{only}}$. Thus $78 - 38 = 44 - 4$. It does not decide which scoring convention best represents the task.

For P9, conditional exact McNemar with $n_d = b+c > 0$ discordances has minimum two-sided value $\min(1, 2^{1-n_d})$, attained when all discordances point one way. Proof: the smallest nonempty binomial tail under probability $1/2$ is 2^{-n_d}; double it. Thus this procedure requires at least six discordances for $p < 0.05$ and eight for $p < 0.0125$. Four

planner gains give $p = 0.125$; five MAS prompt-bundle gains give $p = 0.0625$. This is *not* a claim that no other test could reject, nor evidence of no effect.

For P10, define the recorded cluster score statistic $Q = (\sum_g S_g)^2 / \sum_g S_g^2$, where S_g sums paired event differences and the denominator is nonzero. Cauchy–Schwarz gives $Q \le G'$ for G' nonzero cluster sums. Under the particular asymptotic χ^2_1 reference, its computed tail is at least $\Pr(\chi^2_1 \ge G')$. For $G' = 33$ this is approximately 9.2×10^{-9}. The inequality bounds this statistic and reference tail only. The old claim that *no valid cluster test* can provide smaller p-values is false: other statistics, reference distributions and assumptions have different attainable values. The archived code calls this statistic "Durkalski"; here it is called the *cluster score statistic* and its formula is supplied without claiming every variant in that literature is equivalent.

K.6 Nonidentification and error propagation

Across tiers, RAM, checkpoints, precision, adaptation, prompts, runtime and tasks vary together. Any observed outcomes can be matched by multiple response functions assigning different effects to RAM; the effect of changing RAM alone is unidentified without additional controlled comparisons or modeling assumptions. The same is true of a universal architecture effect. Within Study A, the bottleneck is descriptively located before delivery, but its cause can involve prompts, representation, role allocation, source suggestions or model capability.

A message-only Markov chain/data-processing account would require downstream access to evidence only through upstream messages. Here every role also sees source evidence. Correlated errors and poor verification are observed, but a Markov cascade, an information-loss rate or a causal coordination penalty is not estimated. Stage durations quantify observed work; they do not identify the counterfactual cost of removing a role while preserving capability.

L Statistical Derivations and Complete Numerical Battery

L.1 Paired estimands and small-discordance inference (J1–J4)

For the paired-proportion setting of McNemar [1947], let $d_i = y_{Ri} - y_{Li} \in \{-1, 0, 1\}$ and $\widehat{\Delta} = N^{-1} \sum_i d_i = (b - c)/N$. Under independent case pairs,

$$\mathrm{Var}(\widehat{\Delta}) = \{\mathrm{Var}(y_R) + \mathrm{Var}(y_L) - 2\,\mathrm{Cov}(y_R, y_L)\}/N. \tag{10}$$

Dropping the covariance by treating arms as independent changes the sampling model. Under the multinomial paired model, a plug-in variance is $\{(b + c) - (b - c)^2/N\}/N^2$. The resulting Wald interval can misbehave near boundaries; it is not the primary uncertainty interval.

Conditional on $n_d = b + c$, the null of equal event probabilities gives $b \sim \mathrm{Binomial}(n_d, 1/2)$, and

$$p_{\mathrm{exact}} = \min\left(1, 2 \sum_{k=0}^{\min(b,c)} \binom{n_d}{k} 2^{-n_d}\right). \tag{11}$$

This exactness is conditional on independent/exchangeable pairs under the binomial model; it does not survive arbitrary within-report dependence. The asymptotic statistic is $(b - c)^2/n_d$; the implementation's corrected version is $\max(|b - c| - 1, 0)^2/n_d$. Both use χ^2_1 approximations. The maximum with zero is required to match the code when $b = c$. At zero discordance we conventionally report $p = 1$ with undefined chi-square statistics, not equivalence.

Exact conditional tests can be conservative [Fagerland et al., 2013]. Retaining them alongside bootstrap intervals exposes sparse data rather than choosing the favorable procedure. With five gains and no losses among 160 cases the Wald interval excludes zero but exact $p = 0.0625$; with four gains among 96, the percentile bootstrap can also exclude zero. Neither discrepancy licenses a confident population improvement. There is no post-hoc observed-power calculation.

L.2 Transition tables, clustering and bootstrap (J5–J7)

For states C/W/H and paired counts n_{ij}, Bowker symmetry uses $Q_B = \sum_{i<j} (n_{ij} - n_{ji})^2/(n_{ij} + n_{ji})$, omitting empty pairs. Its asymptotic df is the number of nonempty pairs. Stuart–Maxwell tests marginal homogeneity with $\delta_i = n_{i\cdot} - n_{\cdot i}$, $V_{ii} = n_{i\cdot} + n_{\cdot i} - 2n_{ii}$ and $V_{ij} = -(n_{ij} + n_{ji})$. Dropping a redundant category gives $Q_{SM} = \delta^\top V^+ \delta$,

rank df, and undefined statistics at rank zero. These are case-IID, large-sample diagnostics; sparse off-diagonal cells and shared reports limit their tail probabilities even when the numerical statistics are reproduced exactly.

For independent clusters with sums $S_g = \sum_{i \in g} d_i$ and fixed sizes, $\mathrm{Var}(\widehat{\Delta}) = N^{-2} \sum_g \mathrm{Var}(S_g)$. The recorded CR1 estimate is

$$\widehat{V}_{CR1} = \frac{G}{G-1} \frac{1}{N^2} \sum_g (S_g - \widehat{\Delta} n_g)^2. \tag{12}$$

Its normal-tail z test is asymptotic; the $G/(G-1)$ factor does not guarantee small-sample validity. Cluster-robust methods allow within-cluster dependence under appropriate independent-cluster regularity conditions [Cameron and Miller, 2015]. The score statistic of P10 instead uses uncentered squared cluster sums under its zero-mean null. It reduces to the uncorrected McNemar statistic for singleton clusters. Neither procedure is universally finite-sample valid.

The empirical variance ratio $\widehat{V}_{CR1}/\widehat{V}_{IID}$ is a descriptive design effect, not proof that one study has exactly N/DEFF independent observations. The earlier draft also inferred an intraclass correlation from a common-correlation approximation; heterogeneous binary paired differences do not identify such an ICC. We retain the computed design effects, but exclude that unsupported interpretation.

Cluster bootstrap draws G whole reports with replacement, retaining all paired questions, and recomputes $\sum_g S_g^* / \sum_g n_g^*$. It targets a question-weighted ratio under exchangeable independent report sampling with finite moments and nonzero denominator. Equal-report averaging instead estimates $G^{-1} \sum_g S_g/n_g$. Neither is automatically appropriate to every target population. The 10,000-draw NumPy seed-42 implementation uses linear percentiles; source audit Python-indexed quantiles are kept separately. Percentile intervals are approximate and may under-cover with few discordant clusters. If all observed changes have one sign, the bootstrap cannot synthesize unseen opposite changes; it need not touch zero. This corrects the old blanket statement that such intervals always do so.

L.3 Sign procedures, multiplicity and interpretation (J8–J12)

Under independent cluster sums symmetric about zero, conditioning on magnitudes makes the signs exchangeable. Enumerating all sign assignments yields an exact conditional sign-flip test; the recorded implementation enumerates at most 18 nonzero clusters and otherwise uses 65,536 random assignments with $(E+1)/(R+1)$, not zero. The report sign test instead conditions on nonzero sums and assumes independent equiprobable signs, ignoring magnitudes. Zero average effect alone implies neither null. These tests are supplemental, not randomized-treatment inference.

For valid marginal p-values, Holm's sorted adjustment is $\widetilde{p}_{(j)} = \min(1, \max_{\ell \le j}(m - \ell + 1) p_{(\ell)})$ [Holm, 1979]. If the first true null is rejected when m_0 true nulls remain, its threshold is no larger than α/m_0; the union bound gives family error at most α without independent tests. This guarantee requires valid input p-values; it does not repair invalid cluster assumptions. The four original correct/wrong endpoints form the disclosed family; withholding is algebraically determined. Excluding its test from this family is a design choice, not a theorem forbidding testing withholding for another question.

Wilson intervals invert the independent-binomial score inequality. With $z = 1.96$, their endpoints are

$$\frac{\widehat{p} + z^2/(2n) \;\pm\; z\sqrt{\widehat{p}(1-\widehat{p})/n + z^2/(4n^2)}}{1 + z^2/n}. \tag{13}$$

They remain inside [0,1] but do not account for report dependence or differing selected delivery sets. Risk differences are directly interpretable here and remain finite when discordant odds ratios separate ($b = 0$ or $c = 0$). This does not imply that every logistic, linear-probability or cluster-robust regression is invalid; those alternatives require their own specifications and assumptions. None is needed to reproduce these descriptive paired contrasts.

All inferential interpretations depend on a hypothetical exchangeable-case/report population, despite deterministic cohort construction and greedy decoding. There are no seed, checkpoint-family or financial-expert uncertainty estimates. Non-significance does not establish equivalence without a margin and suitable design. A small p-value neither makes a negative result novel nor resolves the correct/wrong/withheld trade-off. The supplemental battery is retained for transparency, not to multiply apparent evidential strength.

L.4 All historical comparisons, both numerical scores

The following tables reproduce all 21 saved comparisons and all 126 C/W/H endpoint rows, including raw-score 8 GB comparisons that the old PDF relegated to CSV. Values are independently recomputed after checking derived outcomes

against the original raw ledgers. The complete JSON additionally retains Wald intervals, both asymptotic McNemar variants, CR1, company score tests, report sign tests and transition matrices; selected primary statistics are also printed above. Repeated arms, questions, reference scores and tests do not increase the independent sample size.

Among 28 reported C/W endpoints in the 8 GB comparison family, five case-IID exact tests are below 0.05 before adjustment; none survives Holm adjustment (minimum about 0.36). The reserved MAS critic wrong-delivery decrease 17 to 9 has unchanged reported correct count 4 to 4 but raw correct decreases 8 to 7. Thus even a favorable error comparison cannot override the both-score no-correct-loss rule. The detailed battery makes this adverse trade-off visible.

16 GB comparison key. 1. Primary: MAS vs SAS
2. Direct control: MAS vs SAS
3. Repair ablation: MAS vs SAS
4. SAS: primary vs direct
5. MAS: primary vs direct
6. SAS: repair vs primary
7. MAS: repair vs primary

Table 16: Complete 16 GB paired endpoint battery. R: reported, X: raw; C/W/H are events. IDs are defined immediately above. All tests here are post-hoc, unadjusted sensitivity analyses; exact tests assume independent pairs, score tests independent clusters and a large-cluster approximation.

ID / score / event	L/R	gain/loss	Δ pp	Report 95%	Exact p	Score p
1 R C	78/5	0/73	-45.62	[-53.90, -36.55]	2.12e-22	6.41e-07
1 R W	48/3	1/46	-28.12	[-37.09, -20.13]	6.82e-13	1.84e-06
1 R H	34/152	118/0	+73.75	[+64.71, +81.93]	6.02e-36	9.24e-09
1 X C	38/5	1/34	-20.62	[-28.26, -13.37]	2.1e-09	4.26e-05
1 X W	88/3	0/85	-53.12	[-61.90, -44.08]	5.17e-26	6.36e-08
1 X H	34/152	118/0	+73.75	[+64.71, +81.93]	6.02e-36	9.24e-09
2 R C	50/0	0/50	-31.25	[-40.12, -22.63]	1.78e-15	5.01e-06
2 R W	66/5	2/63	-38.12	[-47.65, -29.05]	1.16e-16	4.87e-07
2 R H	44/155	112/1	+69.38	[+59.55, +78.77]	2.2e-32	1.3e-08
2 X C	14/0	0/14	-8.75	[-13.29, -4.74]	0.000122	0.000465
2 X W	102/5	2/99	-60.62	[-70.18, -50.63]	4.06e-27	4.19e-08
2 X H	44/155	112/1	+69.38	[+59.55, +78.77]	2.2e-32	1.3e-08
3 R C	79/5	0/74	-46.25	[-54.67, -37.06]	1.06e-22	6.07e-07
3 R W	50/3	1/48	-29.38	[-38.41, -21.38]	1.78e-13	8.35e-07
3 R H	31/152	121/0	+75.62	[+66.67, +83.87]	7.52e-37	6.07e-09
3 X C	38/5	1/34	-20.62	[-28.26, -13.37]	2.1e-09	4.26e-05
3 X W	91/3	0/88	-55.00	[-63.93, -45.86]	6.46e-27	3.8e-08
3 X H	31/152	121/0	+75.62	[+66.67, +83.87]	7.52e-37	6.07e-09
4 R C	50/78	32/4	+17.50	[+9.76, +24.55]	1.94e-06	0.000465
4 R W	66/48	9/27	-11.25	[-17.98, -3.90]	0.004	0.007
4 R H	44/34	4/14	-6.25	[-12.57, -0.64]	0.031	0.050
4 X C	14/38	27/3	+15.00	[+8.59, +21.62]	8.43e-06	0.000402
4 X W	102/88	15/29	-8.75	[-17.96, +0.72]	0.049	0.080
4 X H	44/34	4/14	-6.25	[-12.57, -0.64]	0.031	0.050
5 R C	0/5	5/0	+3.12	[+0.00, +6.98]	0.062	0.096
5 R W	5/3	3/5	-1.25	[-5.26, +2.55]	0.727	0.527
5 R H	155/152	5/8	-1.88	[-7.24, +3.43]	0.581	0.491
5 X C	0/5	5/0	+3.12	[+0.00, +6.98]	0.062	0.096
5 X W	5/3	3/5	-1.25	[-5.26, +2.55]	0.727	0.527
5 X H	155/152	5/8	-1.88	[-7.24, +3.43]	0.581	0.491
6 R C	78/79	1/0	+0.62	[+0.00, +2.04]	1.000	0.317
6 R W	48/50	2/0	+1.25	[+0.00, +3.40]	0.500	0.157

ID / score / event	L/R	gain/loss	Δ pp	Report 95%	Exact p	Score p
6 R H	34/31	0/3	-1.88	[-4.35, +0.00]	0.250	0.083
6 X C	38/38	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
6 X W	88/91	3/0	+1.88	[+0.00, +4.35]	0.250	0.083
6 X H	34/31	0/3	-1.88	[-4.35, +0.00]	0.250	0.083
7 R C	5/5	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
7 R W	3/3	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
7 R H	152/152	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
7 X C	5/5	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
7 X W	3/3	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
7 X H	152/152	0/0	+0.00	[+0.00, +0.00]	1.000	1.000

Table 17: 16 GB full-transition diagnostics. Asymptotic case-IID assumptions may fail, especially with sparse off-diagonal cells. Rank-zero statistics are undefined, not evidence of equivalence.

ID / score	Bowker Q	df	p	SM Q	rank	p
1 reported	119.00	3	1.27e-25	118.034	2	2.34e-26
1 raw	119.00	3	1.27e-25	118.040	2	2.33e-26
2 reported	110.06	3	1.06e-23	109.157	2	1.98e-24
2 raw	110.04	3	1.08e-23	109.156	2	1.98e-24
3 reported	122.00	3	2.86e-26	121.033	2	5.22e-27
3 raw	122.00	3	2.86e-26	121.039	2	5.21e-27
4 reported	25.15	3	1.44e-05	22.349	2	1.4e-05
4 raw	23.29	3	3.52e-05	23.149	2	9.4e-06
5 reported	5.50	2	0.064	5.500	2	0.064
5 raw	5.50	2	0.064	5.500	2	0.064
6 reported	3.00	2	0.223	3.000	2	0.223
6 raw	3.00	1	0.083	3.000	1	0.083
7 reported	0.00	0	1.000	–	0	1.000
7 raw	0.00	0	1.000	–	0	1.000

Earlier 8 GB FinQA comparison key. 1. Selection52: baseline MAS vs SAS
2. Selection52: control MAS vs SAS
3. Selection52: crossed MAS vs SAS
4. Reserved49: control MAS vs SAS
5. Reserved49: crossed MAS vs SAS
6. Selection52: crossed vs control MAS
7. Selection52: crossed vs original-critic MAS
8. Reserved49: crossed vs control MAS
9. Selection52: crossed vs control SAS
10. Selection52: crossed vs control SAS (analysis128+demos)
11. Reserved49: crossed vs control SAS
12. Selection52: baseline SAS, analysis128+demos vs direct
13. Selection52: control SAS, analysis128+demos vs direct
14. Selection52: crossed SAS, analysis128+demos vs direct

Table 18: Complete 8 GB paired endpoint battery. R: reported, X: raw; C/W/H are events. IDs are defined immediately above. All tests here are post-hoc, unadjusted sensitivity analyses; exact tests assume independent pairs, score tests independent clusters and a large-cluster approximation.

ID / score / event	L/R	gain/loss	Δ pp	Report 95%	Exact p	Score p
1 R C	8/2	0/6	-11.54	[-23.26, -2.00]	0.031	0.083
1 R W	25/19	8/14	-11.54	[-32.56, +7.69]	0.286	0.257
1 R H	19/31	18/6	+23.08	[+0.00, +44.44]	0.023	0.064
1 X C	10/3	0/7	-13.46	[-24.53, -3.85]	0.016	0.052
1 X W	23/18	9/14	-9.62	[-29.17, +8.33]	0.405	0.297
1 X H	19/31	18/6	+23.08	[+0.00, +44.44]	0.023	0.064
2 R C	8/3	0/5	-9.62	[-18.33, -2.00]	0.062	0.059
2 R W	23/19	4/8	-7.69	[-21.82, +9.30]	0.388	0.371
2 R H	21/30	12/3	+17.31	[-2.08, +34.29]	0.035	0.117
2 X C	10/5	0/5	-9.62	[-18.33, -2.00]	0.062	0.059
2 X W	21/17	4/8	-7.69	[-21.82, +9.30]	0.388	0.371
2 X H	21/30	12/3	+17.31	[-2.08, +34.29]	0.035	0.117
3 R C	10/3	0/7	-13.46	[-28.85, -2.00]	0.016	0.108
3 R W	16/15	6/7	-1.92	[-15.69, +15.26]	1.000	0.808
3 R H	26/34	13/5	+15.38	[-6.98, +34.43]	0.096	0.194
3 X C	12/5	0/7	-13.46	[-28.85, -2.00]	0.016	0.108
3 X W	14/13	6/7	-1.92	[-15.69, +15.26]	1.000	0.808
3 X H	26/34	13/5	+15.38	[-6.98, +34.43]	0.096	0.194
4 R C	8/4	1/5	-8.16	[-16.22, +0.00]	0.219	0.102
4 R W	20/17	6/9	-6.12	[-27.27, +10.91]	0.607	0.513
4 R H	21/28	12/5	+14.29	[+1.72, +31.71]	0.143	0.052
4 X C	14/8	1/7	-12.24	[-20.00, -4.35]	0.070	0.034
4 X W	14/13	6/7	-2.04	[-22.50, +13.73]	1.000	0.819
4 X H	21/28	12/5	+14.29	[+1.72, +31.71]	0.143	0.052
5 R C	9/4	1/6	-10.20	[-20.83, +0.00]	0.125	0.132
5 R W	20/9	3/14	-22.45	[-38.71, -7.50]	0.013	0.022
5 R H	20/36	19/3	+32.65	[+17.39, +46.88]	0.000855	0.009
5 X C	15/7	1/9	-16.33	[-26.15, -5.00]	0.021	0.046
5 X W	14/6	3/11	-16.33	[-31.43, -3.77]	0.057	0.033
5 X H	20/36	19/3	+32.65	[+17.39, +46.88]	0.000855	0.009
6 R C	3/3	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
6 R W	19/15	1/5	-7.69	[-17.74, +2.08]	0.219	0.157
6 R H	30/34	5/1	+7.69	[-2.08, +17.74]	0.219	0.157
6 X C	5/5	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
6 X W	17/13	1/5	-7.69	[-17.74, +2.08]	0.219	0.157
6 X H	30/34	5/1	+7.69	[-2.08, +17.74]	0.219	0.157
7 R C	4/3	0/1	-1.92	[-5.77, +0.00]	1.000	0.317
7 R W	22/15	3/10	-13.46	[-25.00, -1.85]	0.092	0.052
7 R H	26/34	11/3	+15.38	[+2.13, +27.66]	0.057	0.046
7 X C	6/5	0/1	-1.92	[-5.77, +0.00]	1.000	0.317
7 X W	20/13	3/10	-13.46	[-25.00, -1.85]	0.092	0.052
7 X H	26/34	11/3	+15.38	[+2.13, +27.66]	0.057	0.046
8 R C	4/4	0/0	+0.00	[+0.00, +0.00]	1.000	1.000
8 R W	17/9	2/10	-16.33	[-26.23, -4.26]	0.039	0.046
8 R H	28/36	10/2	+16.33	[+4.26, +26.23]	0.039	0.046
8 X C	8/7	1/2	-2.04	[-5.88, +0.00]	1.000	0.317
8 X W	13/6	1/8	-14.29	[-25.49, -2.17]	0.039	0.071
8 X H	28/36	10/2	+16.33	[+4.26, +26.23]	0.039	0.046
9 R C	8/10	2/0	+3.85	[+0.00, +11.54]	0.500	0.317

ID / score / event	L/R	gain/loss	Δ pp	Report 95%	Exact p	Score p
9 R W	23/16	4/11	-13.46	[-26.79, +0.00]	0.118	0.108
9 R H	21/26	9/4	+9.62	[-3.51, +23.21]	0.267	0.197
9 X C	10/12	2/0	+3.85	[+0.00, +11.54]	0.500	0.317
9 X W	21/14	4/11	-13.46	[-26.79, +0.00]	0.118	0.108
9 X H	21/26	9/4	+9.62	[-3.51, +23.21]	0.267	0.197
10 R C	9/9	1/1	+0.00	[+0.00, +0.00]	1.000	1.000
10 R W	20/10	3/13	-19.23	[-38.03, -2.08]	0.021	0.068
10 R H	23/33	13/3	+19.23	[+2.08, +38.03]	0.021	0.068
10 X C	11/10	1/2	-1.92	[-6.25, +0.00]	1.000	0.317
10 X W	18/9	3/12	-17.31	[-34.78, -1.85]	0.035	0.072
10 X H	23/33	13/3	+19.23	[+2.08, +38.03]	0.021	0.068
11 R C	8/9	1/0	+2.04	[+0.00, +5.56]	1.000	0.317
11 R W	20/20	3/3	+0.00	[-10.64, +8.62]	1.000	1.000
11 R H	21/20	3/4	-2.04	[-12.28, +10.42]	1.000	0.739
11 X C	14/15	1/0	+2.04	[+0.00, +5.56]	1.000	0.317
11 X W	14/14	3/3	+0.00	[-10.64, +8.62]	1.000	1.000
11 X H	21/20	3/4	-2.04	[-12.28, +10.42]	1.000	0.739
12 R C	8/9	2/1	+1.92	[-4.26, +8.77]	1.000	0.564
12 R W	25/24	5/6	-1.92	[-12.07, +12.50]	1.000	0.763
12 R H	19/19	6/6	+0.00	[-15.22, +11.27]	1.000	1.000
12 X C	10/10	2/2	+0.00	[-7.41, +7.69]	1.000	1.000
12 X W	23/23	5/5	+0.00	[-10.00, +13.95]	1.000	1.000
12 X H	19/19	6/6	+0.00	[-15.22, +11.27]	1.000	1.000
13 R C	8/9	2/1	+1.92	[-5.41, +10.53]	1.000	0.655
13 R W	23/20	4/7	-5.77	[-18.75, +6.82]	0.549	0.405
13 R H	21/23	6/4	+3.85	[-7.55, +15.91]	0.754	0.527
13 X C	10/11	2/1	+1.92	[-5.41, +10.53]	1.000	0.655
13 X W	21/18	4/7	-5.77	[-18.75, +6.82]	0.549	0.405
13 X H	21/23	6/4	+3.85	[-7.55, +15.91]	0.754	0.527
14 R C	10/9	0/1	-1.92	[-6.25, +0.00]	1.000	0.317
14 R W	16/10	2/8	-11.54	[-27.27, +0.00]	0.109	0.134
14 R H	26/33	8/1	+13.46	[+2.17, +28.26]	0.039	0.071
14 X C	12/10	0/2	-3.85	[-9.26, +0.00]	0.500	0.157
14 X W	14/9	2/7	-9.62	[-22.73, +0.00]	0.180	0.132
14 X H	26/33	8/1	+13.46	[+2.17, +28.26]	0.039	0.071

Table 19: 8 GB full-transition diagnostics. Asymptotic case-IID assumptions may fail, especially with sparse off-diagonal cells. Rank-zero statistics are undefined, not evidence of equivalence.

ID / score	Bowker Q	df	p	SM Q	rank	p
1 reported	9.20	3	0.027	9.000	2	0.011
1 raw	10.20	3	0.017	9.947	2	0.007
2 reported	7.27	3	0.064	7.119	2	0.028
2 raw	7.27	3	0.064	7.119	2	0.028
3 reported	7.33	3	0.062	7.311	2	0.026
3 raw	7.33	3	0.062	7.311	2	0.026
4 reported	4.92	3	0.178	3.977	2	0.137
4 raw	5.48	3	0.140	5.000	2	0.082
5 reported	12.67	3	0.005	11.712	2	0.003
5 raw	12.68	3	0.005	11.971	2	0.003
6 reported	2.67	1	0.102	2.667	1	0.102

ID / score	Bowker Q	df	p	SM Q	rank	p
6 raw	2.67	1	0.102	2.667	1	0.102
7 reported	4.77	2	0.092	4.769	2	0.092
7 raw	4.77	2	0.092	4.769	2	0.092
8 reported	5.33	1	0.021	5.333	1	0.021
8 raw	5.78	2	0.056	5.778	2	0.056
9 reported	3.92	2	0.141	3.923	2	0.141
9 raw	3.92	2	0.141	3.923	2	0.141
10 reported	7.40	3	0.060	6.452	2	0.040
10 raw	7.57	3	0.056	6.273	2	0.043
11 reported	1.00	2	0.607	1.000	2	0.607
11 raw	1.00	2	0.607	1.000	2	0.607
12 reported	1.00	3	0.801	0.375	2	0.829
12 raw	1.44	3	0.695	0	2	1.000
13 reported	3.11	3	0.375	0.897	2	0.639
13 raw	3.11	3	0.375	0.897	2	0.639
14 reported	6.44	2	0.040	6.444	2	0.040
14 raw	6.50	3	0.090	6.235	2	0.044

M Worked Examples and Measurement Failures

These examples are purposively selected illustrations, not a frequency estimate. The four original financial cases and the synthetic credit case are restored, plus a score-disagreement example. Programs below are compact renderings of saved traces; they are not invented model quotations. Full recorded responses, scores, source annotations and both historical/Ubuntu records are supplied in `worked_example_traces.json`. Historical per-stage timing is not copied into the Ubuntu interpretation.

M.1 Shared success: ABMD/2007/page_52.pdf-1

The question asks for the possible Impella purchase price including contingent payments. The reference is $56.3 = 11.2+45.1$ million. Catalog values A013=45.1 at text_3 and A017=11.2 at text_4 support both arms' one-step addition. Both executions give 56.3 and both reviewers PASS; both scoring conventions agree. The extractor selected text_3, text_4, text_5 and text_6. This case demonstrates that the team can solve some tasks, not that specialization is superior.

M.2 Evidence found, calculation declined: AAPL/2003/page_24.pdf-1

The question asks for retail-segment capital expenditures before 2003, exclusive of that year's amount. The relevant sentence supplies total 290 and current-year 92, giving 198. SAS executes subtract(A005,A007), obtains 198, receives PASS and is correct under both scores. MAS selects the relevant text_16 along with text_17 and text_18, but its program is

```
{"decision":"INSUFFICIENT","answer_unit":"number","steps":[]}
```

The reviewer returns FAIL and the answer is withheld. Its analysis restates the requested subtraction but does not produce it. This distinguishes evidence access from candidate construction; a causal capacity floor is not identified.

M.3 Executable does not mean meaningful: AAPL/2003/page_24.pdf-2

The question asks for the greatest gross-margin percentage over three years; the reference table maximum is 27.9. SAS produces eight chained subtractions, executing to 6.1 percent, and its 14B reviewer PASSes it. MAS divides A009=27 by A010=2003, executing to approximately 0.0134798, with declared percent units; its reviewer FAILs it. The saved execution value is used here rather than treating the rounded display 0.0135 as exact. A year is used as a numerical operand. Both candidates are wrong, but only SAS delivers. The grammar lacks a direct max operation; the source-member guarantee does not establish semantic suitability.

M.4 Wrong team program accepted: PNC/2018/page_171.pdf-1

The question asks for home-equity lines as a percentage of total credit commitments. SAS calculates $100(16944/181612) = 9.32977997$ percent and receives PASS. It matches the reported 9.3% but not the unnormalized raw execution reference; describing it simply as correct without naming the metric would hide this distinction.

MAS computes $(31/169278) \times 100/169278 \approx 1.08183407 \times 10^{-7}$ with fraction units and receives PASS. Its operands come from recorded source spans, but the calculation answers neither the reference nor a demonstrated alternative financial interpretation. It is wrong under both metrics. This is the single reported-score correct-to-wrong SAS-to-MAS transition.

M.5 A scoring convention changes the label: ABMD/2007/page_52.pdf-2

SAS computes a ratio 0.5948190206 and multiplies by 100, declaring percent units. The resulting 59.48190206 matches the rounded reported label 59.5% after unit normalization, but fails direct comparison to raw reference 0.59482. MAS emits INSUFFICIENT; although its reviewer PASSes that response, invalid execution prevents delivery. Review acceptance is therefore not itself delivery. This example illustrates both scale sensitivity and the deterministic execution guard.

M.6 Synthetic sentinel misclassified as invention: CR-TEST-0001

The credit packet supplies operating profit 52,852,800 PKR and debt service 13,000,000, yielding DSCR 4.0656, and current assets/liabilities 211,000,000/70,000,000, yielding 3.0143 after rounding. SAS emits valid JSON, both calculator citations and the supplied adverse-credit status. It copies the source sentinel NOT_PROVIDED for the last-review date, citing the actual source line, while the output contract requires MISSING. The historical scorer flags this as an invented field (S6). We retain that score but do not reinterpret it as fabricated source content.

MAS's fenced outputs fail the strict JSON contract (M2), and its draft uses nonexistent calculator-style status/date spans. Its saved complete-claim-set status is false and invention field null; the 89/111 aggregate excludes it. This explains why formatting, refusal-contract and grounding errors must not be collapsed into a single factual hallucination rate. The restored examples expose limitations in the measurement as well as in the models.

N Restored Literature Context and Reproducibility Map

FinMAN's financial formula/extraction/calculation decomposition and lightweight checking [Liu et al., 2025] are close antecedents. DCRC combines learned evidence auditing, program synthesis and execution [Chen et al., 2026]. Bench360 measures task/system trade-offs across engines, hardware and quantization choices [Stuhlmann et al., 2026]. These motivate the recipe comparison but do not supply performance measurements for our settings. The main paper also discusses program reasoning, tool-assisted correction, selective prediction, debate, equal-budget comparisons and multiagent failure taxonomies. Positive small-agent collaboration under tool access [Żywot et al., 2026] does not conflict with our bounded negative result. Weight quantization methods such as AWQ [Lin et al., 2026] address compression; our GGUF recipe is not an AWQ implementation. Quantized-agent process failures can change even when headline rewards remain similar [Jang et al., 2026]; we do not import that paper's error rates or attribute our errors to quantization alone. The study does not claim to be the first system with provenance, role specialization, executable arithmetic, repair or abstention.

Peripheral surveys of medical collaboration, cooking benchmarks, quantum-security agents and broad regulatory instruments are not restored as separate subsections: they do not validate these financial experiments. In particular, an assertion that named regulations universally require on-premises inference is not supported here. Unverified venue/version claims and external percentage gains are not copied. The coverage audit identifies these exclusions, and the original bibliography remains preserved; a shorter bibliography is not presented as an exhaustive systematic review.

The reproducibility hierarchy is: frozen raw packets and case ledgers; independent program execution/scoring; reconciliation with derived case tables; recomputed statistical tables; manuscript figures and numerical text. Hashes of original manuscripts are in `../preserved_versions.json`; restoration input fingerprints and checks are in `coverage_evidence_checks.json`. Original complete audit records and evaluator dependencies remain in `../../output/final_aaai/`. The handoff restore utility reconstructs the preserved evidence tree without model

inference. Original unavailable SSD export/log artifacts are still unavailable; new documentation is not a substitute for those missing bytes.

```
python scripts/reporting/coverage_evidence.py --evidence PATH
python -m unittest tests.test_reporting_statistics
python -m unittest tests.test_paper_statistical_tests
```

Terminology: SAS uses one model instance for potentially several stages; MAS has separate resident role instances. Q4_K_M names the GGUF weight quantization, not a four-bit activation or KV-cache claim. F16 names stored half-precision weights. RSS denotes resident-set accounting; Windows commit is committed virtual-memory accounting and is not a physical-RAM guarantee. Coverage is $(C + W)/N$; selective error is $W/(C + W)$ when any answers are delivered; pp denotes percentage points. Historical KYC/credit tasks test synthetic output contracts, not professional due diligence. No 32 GB measurement, general scaling curve or autonomous financial-decision validation is supplied.

Automated assistance and author responsibility

This draft was prepared with AI-assisted manuscript adaptation, code-based numerical checking and plotting using the preserved research record. Human authors must review the scientific text, establish authorship and complete applicable disclosures before submission. No new research-model inference was performed during manuscript preparation.

References

A. Colin Cameron and Douglas L. Miller. A practitioner's guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317.

Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, and Ion Stoica. Why Do Multi-Agent LLM Systems Fail?, 2025. URL https://arxiv.org/abs/2503.13657.

Hao Chen, Xing Tang, Qirui Liu, Weijie Shi, Shiwei Li, Fuyuan Lyu, Weihong Luo, Xiku Du, and Xiuqiang He. Fighting Numerical Hallucinations via Data-centric Compilation for Online Financial QA, 2026. URL https://arxiv.org/abs/2605.31064.

Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. Program of Thoughts Prompting: Disentangling Computation from Reasoning for Numerical Reasoning Tasks, 2023. URL https://arxiv.org/abs/2211.12588.

Zhiyu Chen, Wenhu Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. FinQA: A dataset of numerical reasoning over financial data. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3697–3711, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.300. URL https://aclanthology.org/2021.emnlp-main.300/.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient Finetuning of Quantized LLMs, 2023. URL https://arxiv.org/abs/2305.14314.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 11733–11763. PMLR, 21–27 Jul 2024. URL https://proceedings.mlr.press/v235/du24e.html.

Morten W. Fagerland, Stian Lydersen, and Petter Laake. The McNemar test for binary matched-pairs data: Mid-p and asymptotic are better than exact conditional. *BMC Medical Research Methodology*, 13:91, 2013. doi: 10.1186/1471-2288-13-91.

Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 10764–10799. PMLR, 23–29 Jul 2023. URL https://proceedings.mlr.press/v202/gao23f.html.

Yonatan Geifman and Ran El-Yaniv. SelectiveNet: A deep neural network with an integrated reject option. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 2151–2159. PMLR, 09–15 Jun 2019. URL https://proceedings.mlr.press/v97/geifman19a.html.

Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large Language Models Can Self-Correct with Tool-Interactive Critiquing, 2024. URL https://arxiv.org/abs/2305.11738.

Bethel Hall, Sachi Shome, and William Eiers. VeriFin: A Neurosymbolic Framework for Verifying LLM-Generated Financial Claims, 2026. URL https://arxiv.org/abs/2608.10213.

Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL https://www.jstor.org/stable/4615733.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models, 2021. URL https://arxiv.org/abs/2106.09685.

Jie Huang, Xinyun Chen, Swaroop Mishra, Huaixiu Steven Zheng, Adams Wei Yu, Xinying Song, and Denny Zhou. Large Language Models Cannot Self-Correct Reasoning Yet, 2024. URL https://arxiv.org/abs/2310.01798.

Jiwon Jang, Kisu Yang, Heuiseok Lim, and Hyunwoo Park. Flat Score, Amplified Failures: How the Error Budget Masks Damage in Quantized LLM Agents, 2026. URL https://arxiv.org/abs/2607.27275.

Janet Jenq and Hongda Shen. BudgetVerify: Budget-Tiered Verification for Financial QA, 2026. URL https://arxiv.org/abs/2609.33052.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration, 2026. URL https://arxiv.org/abs/2306.00978.

Chenghao Liu, Qian Liu, Ziqin Zhu, Hao Fei, and Aniket Mahanti. David vs. goliath: Cost-efficient financial QA via cascaded multi-agent reasoning. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 4212–4229, Suzhou, China, November 2025. Association for Computational Linguistics. ISBN 979-8-89176-335-7. doi: 10.18653/v1/2025.findings-emnlp.225. URL https://aclanthology.org/2025.findings-emnlp.225/.

Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative Refinement with Self-Feedback, 2023. URL https://arxiv.org/abs/2303.17651.

Quinn McNemar. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157, 1947. doi: 10.1007/BF02295996.

Qwen Team. Qwen2.5 Technical Report, 2024. URL https://arxiv.org/abs/2412.15115v1.

Linus Stuhlmann, Mauricio Fadel Argerich, and Jonathan Fürst. Bench360: Benchmarking Local LLM Inference from 360 Degrees, 2026. URL https://arxiv.org/abs/2511.16682.

Dat Tran and Douwe Kiela. Single-Agent LLMs Outperform Multi-Agent Systems on Multi-Hop Reasoning Under Equal Thinking Token Budgets, 2026. URL https://arxiv.org/abs/2604.02460.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-Consistency Improves Chain of Thought Reasoning in Language Models, 2023. URL https://arxiv.org/abs/2203.11171.

Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli Feng, and Tat-Seng Chua. TAT-QA: A question answering benchmark on a hybrid of tabular and textual content in finance. In Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli, editors, *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3277–3287, Online, August 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.acl-long.254. URL https://aclanthology.org/2021.acl-long.254/.

Agata Żywot, Xinyi Chen, Yifei Yuan, Anders Søgaard, and Maarten de Rijke. Can Small Agents Collaborate to Beat a Single Large Language Model?, 2026. URL https://arxiv.org/abs/2601.11327.